

ProtogaLattice: Lattice-based Algebraic Folding

David Balbás¹, Anca Nitulescu², and Maxime Plançon³

¹ ETH Zürich, Zurich, Switzerland

² Input Output, France

³ Input Output, Switzerland

Abstract. Folding schemes are gaining traction recently as they are suited to prove streaming computations with low memory footprint. In particular, there has been a growing interest in post-quantum folding schemes for more expressive relations, and with improved proof sizes. While the landscape is vast, every lattice-based construction, such as Latticefold+, (Super)Neo, and Cyclo, heavily rely on the sumcheck protocol. Sumcheck gives efficient proving times, but the verifier circuits become very large, partially because of the many random oracle invocations required. When folding sequential computations, this significantly inflates the recursive overhead, as the prover must run the verification at every iteration.

We present ProtogaLattice, a new lattice-based folding scheme for general high-degree polynomial relations that drastically reduces the size of the verifier’s circuits. Our folding backbone deviates from the sumcheck approach and uses algebraic techniques. Our contribution is twofold:

- i) a folding scheme that combines multiple instances of polynomial relations into accumulators for a bounded number of iterations, and
- ii) a bootstrapping protocol to reduce the norm of the witnesses underlying these accumulators, which can be used in complement to our bounded-depth folding scheme to unlock unlimited depth.

A full iteration of ProtogaLattice for witnesses of length 2^{30} gives a $4\times$ shorter proof size than the state-of-the-art, with only 8.13 KB. Additionally, ProtogaLattice requires less random oracle evaluations and achieves a smaller verifier circuit than its sumcheck counterparts.

Our techniques open new directions towards building efficient lattice-based proofs for expressive relations and that present small recursion overheads, challenging the established sumcheck-based approaches.

1 Introduction

Folding Schemes, first introduced by Nova [KST22], allow one to accumulate multiple instances of an NP relation into a single small instance while preserving satisfiability. As a better alternative to classical recursive proof systems, such techniques unlock efficient instantiations of incrementally verifiable computation (IVC) [Val08] or proof carrying data (PCD) [CT10].

Works that build recursive proof schemes [BCMS20, BCL⁺21] show how to amortize the recursive costs by aggregating some parts of the proof, while the recursive statement only checks that the aggregation was done correctly. These ideas were further developed into more efficient ways to accumulate statements and defer hard computations, which led to the emergence of multiple folding schemes.

Recently, numerous works have extended the folding paradigm, improving efficiency, generality, and modularity.

In a nutshell, folding schemes allow one to aggregate multiple instances of a computational relation into a single instance that captures their joint validity via the *reduction of knowledge* abstraction [KP23]. To illustrate reductions of knowledge, let Ξ and Ξ^* be two instance–witness relations. We denote instances by $\mathbf{t}$ and their corresponding witnesses by $\mathbf{w}$. A reduction of knowledge from Ξ to Ξ^* is an interactive protocol Π between a prover and a verifier with the following

structure: the verifier takes as input an instance $\mathbf{t}_1$ for Ξ , interacts with the prover, and outputs an instance $\mathbf{t}_2$ for Ξ^* . The security property of reductions of knowledge ensures that if the prover can subsequently produce a valid witness $\mathbf{w}_2$ (such that $(\mathbf{t}_2, \mathbf{w}_2) \in \Xi^*$), then there exists an efficient extractor that recovers a valid witness $\mathbf{w}_1$ (such that $(\mathbf{t}_1, \mathbf{w}_1) \in \Xi$).

Then, a folding scheme for k witnesses is simply a reduction of knowledge tailored to accumulations, which goes from a *product relation* $\Xi^* \times \Xi^k \rightarrow \Xi^*$. Here, Ξ is the base relation and Ξ^* is an *accumulation relation* whose instance compactly represent prior computation steps.

To construct IVC, the prover can fold a sequence of computation steps into a single accumulator instance whose size remains essentially fixed, which leads to memory-efficient proving. At the end of the computation, it suffices to prove knowledge of a witness for the final Ξ^* instance.

Folding schemes are characterized by their expressiveness, which is essentially the class of relations that they can fold. Given a witness $\mathbf{w}$ and an instance $\mathbf{t}$, a general relation takes the form

$$\Xi = \{(\mathbf{t}, \mathbf{w}) : f(\mathbf{w}) = \mathbf{0} \wedge \mathbf{t} = \text{cm}(\mathbf{w})\},$$

where f is a polynomial map of total degree at most d , and cm is an additive homomorphic commitment scheme. Depending on the form of f , a folding scheme may capture standard constraint systems used in succinct arguments such as R1CS, Plonkish, CCS, or even arbitrary polynomial constraints. Generally, supporting large d and arbitrary polynomial maps f is preferred as it enables more efficient arithmetizations, improving the practicality.

Without aiming to be exhaustive⁴, we categorize folding schemes in two big families:

Algebraic folding schemes define the folding step as a closed algebraic transformation on the relation itself, so each fold is implemented by evaluating and linearly combining polynomial constraints on the relation witnesses directly. Then, the verifier combines the instances leveraging an additive homomorphic commitment scheme. Examples include Nova [KST22] and FLIP-and-Prove [NPR26] for quadratic relations, and Protostar [BC23] and Protogalaxy [EG23] for high-degree, arbitrary polynomial relations.

Sumcheck-based folding schemes such as Hypernova [KS24] or FLI [GM24] phrase the folding as a claim about a big polynomial (or about products of multilinear polynomials) and then use the sumcheck protocol to reduce the claim to a small number of scalar checks, so the heavy lifting is in the interactive polynomial reduction rather than in an algebraic gadget. This is also the approach taken by all existing lattice-based folding schemes⁵ including Latticefold [BC25a], Latticefold+ [BC25b], Neo/SuperNeo [NS26], and Cyclo [GLLO26] for high-degree customizable constraint systems (CCS).

None of the above approaches is strictly better than the other. On the one hand, algebraic schemes give very structured accumulators, are more succinct, and, in general, have cheaper per-step verification (partially because the round-efficiency, with few random oracle calls), resulting in smaller recursive overheads for IVC. On the other hand, sumcheck-based schemes enjoy linear proving time in k for multi-folding ($k > 1$).

⁴ In particular, for readability we are omitting families of folding schemes that do not rely on additively homomorphic commitments such as [BMNW25,BCFW25]

⁵ Except for Lova [FKNP24], an algebraic folding scheme for the subset sum problem over the integers.

1.1 Lattice-based Folding

Due to their plausible post-quantum security and to the capacity of supporting smaller (64-bit) fields natively, lattice-based folding schemes have seen lots of research activity in recent years.⁶ All existing lattice-based folding schemes for high-degree relations work over rings of the form $\mathcal{R}_q = \mathbb{Z}_q[X]/(\Phi)$ for a cyclotomic polynomial $\Phi = X^N + 1$ and follow the same high-level recipe:

1. Depart from a high degree- d relation that can be captured by products of multilinear polynomials, such as CCS [STW23]. Define witnesses $\mathbf{w}_i$ for these relations with low norm, that is, $\|\mathbf{w}_i\| \leq B$ for some bound B .
2. Commit to the $\mathbf{w}_i$'s to be folded using Ajtai commitments [Ajt96], which are additively homomorphic and binding for low-norm committed vectors. Besides, use a range proof to show that the committed $\mathbf{w}_i$'s satisfy the norm bound $\|\mathbf{w}_i\| \leq B$.
3. Reduce, via the sumcheck protocol, the initial degree- d relation to a series of evaluation claims about *multilinear* (i.e. degree-1) polynomials. These claims are generally captured as a *linearized relation* (e.g. Linearized CCS).
4. Decompose the witnesses in some basis b to control the norm growth induced by the folding as needed.

This elegant approach has been proven to be successful, but it is currently the only recipe for building lattice-based folding schemes with practical performance and which are expressive enough to capture high-degree relations. Besides, it presents two important limitations, which seem very intrinsic to the reliance on sumchecks. First, the high-degree relation used needs to be sumcheck-friendly. This prevents one from folding arbitrary transcripts determined by polynomial constraints, such as in algebraic folding schemes [BC23,EG23], which are more versatile and enable native folding of Plonkish and lookup relations without translation overheads. Second, sumcheck leads to large transcripts, many random oracle calls (e.g. Latticefold+ reports around 100 RO calls), and therefore a complex verifier circuit. When used in an IVC construction, all verifier operations, including evaluations of hash functions, have to be encoded in the prover circuit, which is a significant source of inefficiency.

Considering the state-of-the art in the lattice-based folding schemes, our work addresses the following question:

Can we build a lattice-based algebraic folding scheme that:

- (i) *is round-efficient,*
- (ii) *communication-efficient,*
- (iii) *expressive for high-degree relations,*
- (iv) *and supports an unbounded number of folding iterations?*

1.2 Our Contributions

We introduce ProtogaLattice, a new algebraic folding paradigm for folding arbitrary polynomial relations from lattice-based assumptions. Our new algebraic folding techniques offer small proofs,

⁶ We remark that lattice-based folding techniques have also been applied to realize other families of constructions, such as SNARKs [BS23,KLNO24,KLNO25,NDC⁺24,Che25,KLN⁺26], polynomial commitments [AFLN24,CMNW24,NS24], and generally, range proofs.

low verifier circuit complexity and the ability to capture general relations (arbitrary-degree polynomial constraints) natively. These relations are defined by transcripts of special-sound protocols, like in [BC23,EG23], as they efficiently capture R1CS, CCS, Plonkish or lookup arithmetizations. We also present a novel method for norm-control, a *witness bootstrapping* protocol, that replaces the witness decomposition needed in sumcheck folding schemes. Our bootstrapping tool can be used complementary to our (bounded-norm) folding to unlock unbounded-depth folding composition for IVC applications.

Overall, our work displays a fresh, unexplored route towards building efficient lattice-based folding, challenging the assumption that sumcheck-based techniques are the only viable approach. Concretely, our contributions can be split into two main, independent protocols which we describe below.

A lattice-based folding scheme for polynomial relations. Our first and main result, which we introduce in Section 5.2, is a folding protocol for arbitrary polynomial relations defined over the ring $\mathcal{R}_q = \mathbb{Z}_q[X]/(X^N + 1)$ for a power-of-two N . Our folding protocol is a reduction of knowledge for relations $\Xi^{\text{acc}} \times (\Xi^{\text{fold}})^k \rightarrow \Xi^{\text{acc}}$. Here, Ξ^{acc} is a relaxed relation for accumulators of bounded norm, and Ξ^{fold} is the relation for fresh witnesses of bounded norm.

Our starting point is an attempt to carry out the techniques of algebraic folding schemes over groups, such as Protogalaxy [EG23], to the ring setting. In Protogalaxy, k witnesses $\mathbf{w}_1, \dots, \mathbf{w}_k$ are linearly combined into an accumulator $\mathbf{w}^*$ using (univariate) Lagrange polynomials over a domain $\mathbb{H}$ of size k . Soundness and completeness rely on algebraic relations modulo the vanishing polynomial of $\mathbb{H}$. Translating this to lattices, however, fails due to the norm expansion of the accumulator $\mathbf{w}^*$ that results from evaluating the Lagrange polynomials. In particular, when using Ajtai commitments [Ajt96] where $\mathbf{t} = \mathbf{A}\mathbf{w} \pmod{q}$ is a homomorphic commitment to a small $\mathbf{w} \in \mathcal{R}_q^m$ for a public matrix $\mathbf{A} \in \mathcal{R}^{\kappa \times m}$, the commitment is no longer binding for $\mathbf{w}$ with large norm. As a workaround, we combine witnesses using *multivariate* polynomials instead. For this to work, we develop an analogous algebraic machinery using ideals over the ring $\mathcal{R}_q[X_1, \dots, X_k]$ and the corresponding Gröbner basis theory. We provide a more detailed technical overview in Section 5.1.

This gives us a *relaxed folding* scheme where the commitments satisfy only a relaxed binding property. For soundness to hold, our scheme requires the prover to provide a *folding range proof* for the fresh input witnesses (not on the accumulator) as shown in Cyclo [GLLO26].

We present our folding scheme as a modular framework composed by a *relaxed folding* protocol and a *folding range proof*, that we model as reductions of knowledge [KP23]. We further instantiate our folding range proof by adapting the sumcheck-based ternary proof from Neo and SuperNeo [NS26]. This framework admits a broad class of folding range proofs, so any future improvement on these schemes for lattices, such as an algebraic range proof with fewer rounds, could be plugged in to reduce the verifier circuit of ProtogaLattice even further.

Norm control via witness bootstrapping. Our folding scheme can be iterated multiple times (we set parameters to support $L = 64$ iterations), but eventually the norm of the accumulator grows too large to continue folding. Our second result is a bootstrapping protocol for reducing the norm of accumulators obtained from folding. This protocol can be used after many folding iterations, when the norm reaches the upper bound, to reset the running accumulator before continuing with more folding steps. Concretely, our bootstrapping protocol is a reduction of knowledge from $\Xi_{B'}^{\text{acc}} \rightarrow \Xi_B^{\text{acc}}$, where the output norm bound B is smaller than the input norm bound B' .

Efficiency evaluation. We also adopt a concrete set of parameters and evaluate our scheme for higher-degree relations and multiple instances to fold, for which we show the corresponding proof sizes in Table 1. A detailed discussion of the performance and more estimates can be found in Section 6.

Witness Size	Proof Size (KB)					
	deg(f) = 2			deg(f) = 5		
	$k = 1$	$k = 2$	$k = 3$	$k = 1$	$k = 2$	$k = 3$
0.1 GiB ($\sim 2^{30}$)	8.13	14.44	25.36	8.88	27.09	110.86
1.0 GiB ($\sim 2^{33}$)	9.03	17.05	29.03	9.78	30.55	118.07

Table 1. Concrete proof sizes of ProtogaLattice for a folding depth $L = 64$ and k fresh instances to fold for a relation defined by the polynomial-map f . We exclude the commitments and the instances and report only the proof, in line with [NS26, GLLO26, Osa26]. See Table 5 for more data.

1.3 Concrete Efficiency and Related Work

In Table 2, we provide a comparison between ProtogaLattice and prior lattice-based folding schemes for high-degree relations. We remark that while all these schemes support high-degree relations, they all provide performance numbers for quadratic relations (e.g. R1CS). Hence, for a fair comparison we also report efficiency numbers for folding one instance at a time for a degree-2 relation with ProtogaLattice.

	SuperNeo [NS26]	Latticefold+ [BC25a]	Cyclo [GLLO26]	ProtogaLattice (This paper)
Folded relation	CCS/LCCS	CCS/LCCS	CCS/LCCS	Polynomial relations (incl. CCS)
Techniques	HyperNova style sumcheck	HyperNova style sumcheck	HyperNova style sumcheck	Multivariate polynomial interpolation
Range proof	Binary sumcheck proof	Wider proof, uses ring monomials	Ternary sumcheck proof	Ternary sumcheck proof
Folding Depth	∞	∞	64 bounded	64 ∞ with bootstrap
Proof size ($ witness = 2^{27}$)	~ 100 KB	~ 100 KB	31.4 KB	6.52 KB
# Rounds	> 60	~ 100	~ 60	~ 30

Table 2. Comparison table between ProtogaLattice and prior lattice-based folding schemes for high-degree relations. We consider degree 2 relations (R1CS), witness size of total integer length 2^{27} and only folding $k = 1$ fresh instance at each step.

Concurrent work. PikkuFold [Osa26] advances the line of sumcheck-based folding schemes by integrating optimizations—such as challenge rejection sampling—that are also applicable to our construction. Its primary departure from prior work is its range proof module, which is designed to minimize the prover-to-verifier communication. While this range proof module could fit our framework as well, it requires the verifier to operate on $\mathcal{O}(\lambda^2 \log m)$ bits of projection matrices, which we estimate is at least tens of megabytes in practice. While generating such large matrices can be implemented very efficiently, the translation into constraints of such verifier circuit is presumably complex and adds significant overheads in the iterative setting.

1.4 Summary and Open Problems

Summary. In this work, we construct ProtogaLattice, a novel lattice-based folding scheme for high-degree polynomial relations that is round and communication efficient and allows for unbounded folding depth. ProtogaLattice supports folding k witnesses at a time (although efficiency degrades significantly for large k), and works by iteratively folding and bootstrapping witnesses to reduce their norm.

Our construction can be seen as a constant-round folding backbone to which an efficient range-proof mechanism can be attached, making it modular and flexible for instantiations. This represents a significant step forward in the design of low-round folding schemes for general relations.

Open Problems. While ProtogaLattice shows a novel approach towards realizing efficient lattice-based folding for expressive general relations, there are some open questions that we consider interesting and leave as future work.

1. *Constant-Round Norm Check.* ProtogaLattice provides a flexible constant-round folding module to which any range proof available in the literature can be attached. We instantiate it with a sumcheck-based range proof adapted from (Super)Neo [NS26]. To the best of our knowledge, succinct range proofs with succinct verification from the literature follow either a split-and-fold paradigm (RoK and Roll [KLNO25], RoK Paper SISsors [KLNO24]) or rely on the sum-check protocol (LatticeFold+ [BC25b], SALSAA [KLOT25], RoKoko [KLN⁺26]). All aforementioned range proofs have a logarithmic number of rounds. Therefore, instantiations using one of these schemes do not allow to fully realize the constant-round potential of the ProtogaLattice folding backbone. We leave as an interesting open question the design of a constant-round range proof that exploits the structure of ProtogaLattice.
2. *Zero-Knowledge.* Depending on the context in which Protogalattice is used, it may be useful to have a zero-knowledge variant of the scheme, or to study whether the scheme (or a modification of it) satisfies the *randomizing* property introduced in [KS24] for the purpose of building zero-knowledge IVC.

2 Preliminaries

2.1 Notation

We denote by $\mathbb{N}$ the set of natural numbers ≥ 0 . We denote the security parameter by $\lambda \in \mathbb{N}$. We call a function ϵ *negligible*, denoted $\epsilon(\lambda) = \text{negl}(\lambda)$, if $\epsilon(\lambda) = O(\lambda^{-c})$ for every constant $c > 0$, and call a function $p(\lambda)$ *polynomial*, denoted $\text{poly}(\lambda)$, if $p(\lambda) = O(\lambda^c)$ for some constant $c > 0$. We say that an algorithm is *probabilistic polynomial time* (PPT) if it consumes randomness and its running

time is bounded by some $p(\lambda) = \text{poly}(\lambda)$. For a finite set S , $x \leftarrow S$ denotes sampling x uniformly at random in S . For an algorithm A , we write $y \leftarrow A(x)$ for the output of A on input x . For two positive $n, m \in \mathbb{N}$, $[n, m]$ is the set $\{n, n+1, \dots, m\}$, and $[n]$ is the set $[1, n]$. We denote vectors and matrices using bold fonts. For a vector $\mathbf{x}$, x_i denotes its i -th component, and for a matrix $\mathbf{M}$, $M_{i,j}$ denotes its (i, j) -th entry. We write $\mathbf{0}_k$ and $\mathbf{1}_k$ for the vectors of k zeros and ones respectively. We also write $\mathbf{0}_{k \times k}$ and $\mathbf{I}_{k \times k}$ for the zero matrices and the identity matrices of dimension $k \times k$ respectively. Given two strings x, y , we denote their concatenation by $x|y$. We use the notation Ξ to denote an efficiently decidable binary relation. For pairs $(\mathbf{x}, \mathbf{w}) \in \Xi$ we call $\mathbf{x}$ the statement and $\mathbf{w}$ the witness. We write $\Xi = \{(\mathbf{x}; \mathbf{w}) : p(\mathbf{x}, \mathbf{w})\}$ to describe an NP relation. For ease in notation, we sometimes include the statement $\mathbf{x}$ as a part of the witness and omit it, i.e. $\Xi = \{\mathbf{w} : p(\mathbf{w})\}$. We denote the infinity norm simply by the operator $\|\cdot\| = \|\cdot\|_\infty$.

2.2 Reductions of Knowledge

We review the *reductions of knowledge* framework [KP23]. We adapt the definition from prior works [NS26, KLNO24] and, as is necessary in the lattice setting, decouple the relations used for completeness from those used for soundness (extraction).

Definition 1 (Reduction of Knowledge). Let Ξ_1, Ξ_2 be relations, called the folding relations, and let Ξ'_1, Ξ'_2 be relations, called the extraction relations, satisfying $\Xi_i \subseteq \Xi'_i$ for $i \in \{1, 2\}$. A reduction of knowledge from $\Xi_1 \rightarrow \Xi_2$ in the completeness direction and $\Xi'_1 \leftarrow \Xi'_2$ in the soundness direction is a tuple of three PPT algorithms $(\mathcal{G}, \mathcal{P}, \mathcal{V})$ defined as follows:

- $\mathcal{G}(1^\lambda, 1^\ell) \rightarrow \mathbf{p}$: Takes a security parameter 1^λ and a size bound 1^ℓ and outputs public parameters $\mathbf{p}$, which we assume are input to all algorithms.
- $\mathcal{P}(t_1, w_1) \rightarrow (t_2, w_2)$: Takes as input an instance-witness pair (t_1, w_1) and interactively reduces the task of checking $(t_1, w_1) \in \Xi_1$ to the task of checking $(t_2, w_2) \in \Xi_2$, or returns $\perp$.
- $\mathcal{V}(\mathbf{p}, t_1) \rightarrow t_2$: Takes as input an instance t_1 and interactively reduces the task of checking t_1 to the task of checking a new instance t_2 .

We write $\langle \mathcal{P}, \mathcal{V} \rangle(t_1, w_1) \rightarrow (t_2, w_2)$ to denote the interaction between $\mathcal{P}, \mathcal{V}$. The reduction must satisfy the following properties:

Completeness. For any admissible set of parameters $\lambda, \ell \in \mathbb{N}$, given $\mathbf{p} \leftarrow \mathcal{G}(1^\lambda, 1^\ell)$ and $(t_1, w_1) \leftarrow \mathcal{A}(\mathbf{p})$ such that $(t_1, w_1) \in \Xi_1$, the prover's output instance must be equal to the verifier's output instance t_2 , and $\Pr[\langle \mathcal{P}, \mathcal{V} \rangle(t_1, w_1) \in \Xi_2] = 1$.

Knowledge soundness. For any expected PPT adversary $\mathcal{A}$, there exists an expected poly-time extractor $\mathcal{E}$ with rewinding access to $\mathcal{A}$ such that if

$$\epsilon(\mathcal{A}) := \Pr \left[(t_2, w_2) \in \Xi'_2 \left| \begin{array}{l} \mathbf{p} \leftarrow \mathcal{G}(1^\lambda, 1^\ell) \\ t_1 \leftarrow \mathcal{A}(\mathbf{p}) \\ (t_2, w_2) \leftarrow \langle \mathcal{A}, \mathcal{V} \rangle(t_1) \end{array} \right. \right] \geq \frac{1}{\text{poly}(\lambda)},$$

then

$$\Pr \left[(t_1, w_1) \in \Xi'_1 \left| \begin{array}{l} \mathbf{p} \leftarrow \mathcal{G}(1^\lambda, 1^\ell) \\ t_1 \leftarrow \mathcal{A}(\mathbf{p}) \\ w_1 \leftarrow \mathcal{E}^{\mathcal{A}}(\mathbf{p}, t_1) \end{array} \right. \right] \geq \epsilon(\mathcal{A}) - \text{negl}(\lambda).$$

Public coin. All verifier messages are bounded-length random messages, and the verifier does not use any other random coins.

2.3 Cyclotomic rings

Let $N \in \mathbb{N}$ be a power of two. We denote by $\mathcal{R} := \mathbb{Z}[X]/\langle X^N + 1 \rangle$ the power-of-two cyclotomic ring of dimension N . Let $q > 2$ be a prime and denote by $\mathcal{R}_q := \mathcal{R}/q\mathcal{R} = \mathbb{Z}_q[X]/\langle X^N + 1 \rangle$ where we represent $\mathbb{Z}_q := \{-\lfloor q/2 \rfloor, \dots, \lfloor q/2 \rfloor\}$. If $q \equiv 1 + 2d' \pmod{4d'}$ for some $d' \mid N$, it is well-known that $\mathcal{R}_q \cong \mathbb{F}_{q^{N/d'}}$ via the Number Theoretic Transform (NTT).

For $f = \sum_{i \in [N]} f_i X^i \in \mathcal{R}_q$, we use $\text{cf}(f) := (f_0, \dots, f_{N-1}) \in \mathbb{Z}_q^N$ to denote the coefficient vector of f , and $\text{ct}(f) := f_0$ is the constant term of f . For a vector $\mathbf{f} \in \mathcal{R}_q^n$, we denote by $\text{cf}(\mathbf{f}) := (\text{cf}(\mathbf{f}_0)^\top, \dots, \text{cf}(\mathbf{f}_{n-1})^\top) \in \mathbb{Z}_q^{n \times N}$ the concatenation of the transposed coefficient vectors, and $\text{ct}(\mathbf{f}) = (\text{ct}(\mathbf{f}_0), \dots, \text{ct}(\mathbf{f}_{n-1})) \in \mathbb{Z}_q^n$ is the first column of $\text{cf}(\mathbf{f})$.

2.4 Sampling Sets

Definition 2 (Sampling sets, [CKP19]). For an arbitrary ring $\mathcal{R}$, a subset $\mathcal{C}$ of $\mathcal{R}$ is a *sampling set* if the difference of any two distinct elements in $\mathcal{C}$ is not a zero divisor. $\mathcal{C}$ is further a *strong sampling set* if the difference is also invertible.

Example: For $\mathcal{R} := \mathcal{R}_q$ where q is a prime, $\mathbb{Z}_q \subseteq \mathcal{R}_q$ is a strong sampling set as the difference of any two distinct elements in this set is invertible in $\mathcal{R}_q$.

Expansion Factor. To ensure that homomorphic commitments remain binding, we need a strong sampling set $\mathcal{C}_{\text{small}} \subseteq \mathcal{R}_q$ such that for every $c \in \mathcal{C}_{\text{small}}$ and any $\hat{\mathbf{v}} \in \mathcal{R}$, the norm of $c\hat{\mathbf{v}}$ will not increase much compared to $\|\hat{\mathbf{v}}\|_\infty$. To quantify this property, we define the **expansion factor** of $\mathcal{C}_{\text{small}} \subseteq \mathcal{R}_q$ as

$$\|\mathcal{C}_{\text{small}}\|_{\text{op}} := \sup_{c \in \mathcal{C}_{\text{small}}, \hat{\mathbf{v}} \in \mathcal{R}} \frac{\|c \times \hat{\mathbf{v}}\|_\infty}{\|\hat{\mathbf{v}}\|_\infty}. \quad (1)$$

Lemma 1 (Generalized Schwartz-Zippel [BCPS18]). Consider a k -variate nonzero polynomial $f \in \mathcal{R}^{\leq d}[X_1, \dots, X_k]$ over a ring $\mathcal{R}$ with per-variable degree at most d . Let $\mathcal{C} \subseteq \mathcal{R}$ be a sampling set.

Then we have $\Pr_{\mathbf{r} \leftarrow \mathcal{C}^k} [f(\mathbf{r}) = 0] \leq \frac{dk}{|\mathcal{C}|}$.

Lemma 2 (Lyubashevsky–Seiler, power-of-two case). Let N and d' be powers of two with $1 < d' \leq N$, and let q be a prime satisfying $q \equiv 1 + 2d' \pmod{4d'}$. Then in the ring $\mathcal{R}_q = \mathbb{Z}_q[X]/(X^N + 1)$ we have the factorization

$$X^N + 1 = \prod_{i=1}^{d'} (X^{N/d'} - r_i) \pmod{q},$$

for distinct $r_i \in \mathbb{F}_q^\times$, so that each $X^{N/d'} - r_i$ is irreducible mod q .

Moreover, any non-zero $y(X) = \sum_{i=0}^{N-1} y_i X^i \in \mathcal{R}$ is invertible in $\mathcal{R}_q$ if either $\|y\|_\infty < \frac{q^{1/d'}}{\sqrt{d'}}$ or $\|y\|_2 < q^{1/d'}$.

2.5 Module SIS

We recall the Module Short Integer Solution (MSIS) problem defined in [LS15,PR06,LM06,ACK21]. We will focus on the ℓ_∞ -norm.

Definition 3 (Module SIS with ℓ_∞ -norms [ACK21]). Let $R := \mathbb{Z}[X]/(X^N + 1)$ and $\mathcal{R}_q := \mathcal{R}/q\mathcal{R}$. On input $\mathbf{A} \leftarrow \mathcal{R}_q^{\kappa \times m}$, the goal of the $\text{MSIS}_{\kappa,m,B}^{\infty,q}$ problem is to find a non-zero $\mathbf{x} \in \mathcal{R}^m$ such that $\|\mathbf{x}\|_\infty \leq B$ and $\mathbf{A}\mathbf{x} = \mathbf{0}$ over $\mathcal{R}_q$.

With the right choice of parameters, we can obtain a $\text{MSIS}_{\kappa,m,B}^{\infty,q}$ problem with 128 bits of security [ESS⁺19].

2.6 Ajtai Commitments

Let $\mathcal{R} := \mathbb{Z}[X]/(X^N + 1)$ and $\mathcal{R}_q := \mathcal{R}/q\mathcal{R}$ where $q \in \mathbb{N}$ is a prime. The commitment $\text{CM}_{\kappa,m,B}$ introduced by [Ajt96] works as follows:

- $\text{KeyGenCom}(\kappa, m) \rightarrow \mathbf{A}$: sample a random matrix $\mathbf{A} \leftarrow \mathcal{R}_q^{\kappa \times m}$.
- $\text{Com}(\mathbf{A}, \mathbf{x}) \rightarrow \text{cm}$: given $\mathbf{x} \in \mathcal{R}^m$ as input, where $\|\mathbf{x}\|_\infty \leq B$, and no randomness, output $\text{cm}(\mathbf{x}) := \mathbf{A}\mathbf{x} \bmod q \in \mathcal{R}_q^\kappa$.

Assuming that the MSIS problem $\text{MSIS}_{\kappa,m,2B}^{\infty,q}$ is hard it follows that $\text{CM}_{\kappa,m,B}$ satisfies the binding property for inputs $\|\mathbf{x}\|_\infty \leq B$: Any adversary that can open a commitment cm to two different openings $\mathbf{x}_1, \mathbf{x}_2$ (with ℓ_∞ -norm less than B), computes $\mathbf{x}_1 - \mathbf{x}_2 \neq \mathbf{0}$ which is a solution to the $\text{MSIS}_{\kappa,m,2B}^{\infty,q}$ problem where $\|\mathbf{x}_1 - \mathbf{x}_2\|_\infty < 2B$.

In our results, to argue knowledge soundness of the folding schemes based on Ajtai commitments, we need the commitments to satisfy a relaxed binding property as defined in [ACK21,ALS20].

2.7 Relaxed Binding Commitment

Let $\mathcal{C} \subset \mathcal{R}_q$ be a strong sampling set with expansion factor T . We say that the pair $(\Delta, \mathbf{x}) \in (\mathcal{C} - \mathcal{C}) \times \mathcal{R}^m$ is a B -weak opening of cm if $\Delta \cdot \text{cm} = \mathbf{A}\mathbf{x} \bmod q$ and $\|\mathbf{x}\|_\infty \leq B$. Informally, a commitment is B -relaxed binding if it is infeasible to find two different B -weak openings for the same commitment. The Ajtai commitment is B -relaxed binding if $\text{MSIS}_{\kappa,m,4TB}^{\infty,q}$ is hard: an adversary that outputs two B -weak openings $(\Delta_1, \mathbf{x}_1), (\Delta_2, \mathbf{x}_2)$ for the same commitment such that $\Delta_1 \mathbf{x}_2 \neq \Delta_2 \mathbf{x}_1$ breaks the $\text{MSIS}_{\kappa,m,4TB}^{\infty,q}$.

3 Warm-up: Algebraic Folding and Protogalaxy

Before getting into technicalities and the description of ProtogaLattice, we provide some intuition by presenting Protogalaxy, the (classical) algebraic folding scheme that served us as a starting point. We also introduce some notations that will be useful later. Protogalaxy [EG23] is a folding scheme which supports general polynomial relations.⁷ In a nutshell, it can fold k witnesses $\mathbf{w}_1, \dots, \mathbf{w}_k$ that

⁷ Protogalaxy actually supports folding of transcripts of special-sound protocols, and so does our scheme ProtogaLattice. For simplicity, here we introduce Protogalaxy for the case when special-sound protocol transcripts have only one message, such as when folding vanilla R1CS or CCS. We refer to Definition 14 and Section B.1 for details—the extension is fairly straightforward.

satisfy a base polynomial relation defined by

$$\Xi := \left\{ (\mathbf{t}, \mathbf{w}) \mid \mathbf{t} = \text{Com}(\mathbf{w}) \wedge f(\mathbf{w}) = \mathbf{0}_n \right\}. \quad (2)$$

for some polynomial map $f = (f_1, \dots, f_n) : \mathbb{F}^m \rightarrow \mathbb{F}^n$ of degree d defined over a finite field $\mathbb{F}$, and an additively homomorphic commitment scheme $\text{Com}(\cdot)$. Such witnesses are combined with an initial *accumulator* $\mathbf{w}_0$ into a fresh accumulator $\mathbf{w}^*$. Given $\mathbf{w}_0$ and then $\mathbf{w}_1, \dots, \mathbf{w}_k$, the folding scheme works by defining a polynomial

$$W(X) = L_0(X)\mathbf{w}_0 + \sum_{i \in [k]} L_i(X)\mathbf{w}_i$$

where $L_0(X), \dots, L_k(X)$ are a basis of Lagrange polynomials over some domain $\mathbb{H}$ of size $k + 1$. Then, given a random challenge $\gamma \in \mathbb{F}$ sent by the verifier, the prover sets $\mathbf{w}^* = W(\gamma)$ as the new accumulator. This strategy minimizes the number of rounds of interaction between prover and verifier, which, when compiled via Fiat-Shamir, results into fewer random oracle calls and smaller verifier circuits.

For both completeness and soundness, the scheme critically relies on the fact that Lagrange polynomials behave as linear functions modulo $Z(X) = \prod_{a \in \mathbb{H}} (X - a)$, the vanishing polynomial over $\mathbb{H}$. In short, this enables the prover to provide a quotient of $f(W(X))$, which the verifier can validate via a linear check. We state the linearity lemma from [EG23, Lemma 4.2], which we formulate for arbitrary domains:

Lemma 3. *Let $f : \mathbb{F}^m \rightarrow \mathbb{F}$ be a polynomial map and $\mathbf{w}_1, \dots, \mathbf{w}_k \in \mathbb{F}^m$ vectors. Let $(L_i)_{i \in [k]}$ be the Lagrange basis over a domain $\mathbb{H}$ of size k and $Z(X)$ the vanishing polynomial of $\mathbb{H}$. Then, we have that*

$$f \left(\sum_{i=1}^k L_i(X)\mathbf{w}_i \right) = \sum_{i=1}^k L_i(X)f(\mathbf{w}_i) \mod Z(X)$$

Proof. It suffices to show that $Z(X)$ divides the polynomial given by $H(X) = f \left(\sum_{i=1}^k L_i(X)\mathbf{w}_i \right) - \sum_{i=1}^k L_i(X)f(\mathbf{w}_i)$. To see this, note that given any $j \in [k]$, and taking $a_j \in \mathbb{H}$, we have that $L_j(a_j) = 1$ and $L_i(a_j) = 0$ for any $i \in [k], i \neq j$. Therefore,

$$\begin{aligned} H(a_j) &= f \left(\sum_{i=1}^k L_i(a_j)\mathbf{w}_i \right) - \sum_{i=1}^k L_i(a_j)f(\mathbf{w}_i) \\ &= f(L_j(a_j)\mathbf{w}_j) - L_j(a_j)f(\mathbf{w}_j) = 0 \end{aligned}$$

As $H(a_j) = 0$ for every $j \in [k]$, then $Z(X)$ divides $H(X)$ as claimed. $\square$

Accumulators. The relation satisfied by the accumulator is Ξ_{rand} , a relaxed relation. The latter are customary in folding schemes, such as e.g. Relaxed R1CS [KST22,NBS23] or Linearized CCS [KS24,BC25a,NS26], but Protogalaxy has the peculiarity that accumulator relations are also high-degree relations. As opposed to witnesses, for which $f(\mathbf{w}) = \mathbf{0}_n$, accumulators evaluate to a non-zero vector under f .

Formally, Protogalaxy can be stated as a reduction of knowledge $\Xi_{\text{rand}} \times \Xi^k \rightarrow \Xi_{\text{rand}}$. We introduce the accumulator relation below.

Definition 4 (Structured randomized relation). *The generalized randomized relation Ξ_{rand} is the subset $\Xi_{\text{rand}} \subseteq (\mathcal{C} \times \mathbb{F}^t \times \mathbb{F}) \times \mathbb{F}^m$ s.t.*

$$\Xi_{\text{rand}} := \left\{ ((\mathbf{t}, \boldsymbol{\beta}, e), \mathbf{w}) \mid \mathbf{t} = \text{Com}(\mathbf{w}) \wedge \sum_{i=1}^n \text{pow}_i(\boldsymbol{\beta}) f_i(\mathbf{w}) = e \right\}$$

where the polynomials $\text{pow}_i(\boldsymbol{\beta}) := \bigotimes_{i=1}^t (1, \beta_i)$ expand a seed $\boldsymbol{\beta} \in \mathbb{F}^t$ of length $t = \lceil \log n \rceil$.

Note that this relation compresses the “rows” of $f(\mathbf{w})$ using a vector of randomizers $\boldsymbol{\beta} \in \mathbb{F}^t$. To prevent the vector $\boldsymbol{\beta}$ from having size n , which would result in a linear-time verifier, Protogalaxy relies on the power monomials function pow_i . This is a form of compressing $\boldsymbol{\beta}$ into a vector of size $\log n$ which still enables efficient re-randomization. In ProtogaLattice, we use the same randomization structure, except over ring elements.

For completeness, we include the full protocol description from [EG23] in Section B.

4 Relations, Architecture and Range Proofs

In this section, we present the relations and architecture of our folding scheme. The architecture’s formal description is presented at the end of the section in Figure 1. The folding backbone presented in Section 5 only allows the extractor to recover relaxed openings, thus we formally introduce the definition of *Relaxed Folding Scheme* (Definition 8). In Section 4.3 we define the *Folding Range Proof* module that is needed to make ProtogaLattice a tight folding scheme.

4.1 Relations

The two relations in Definition 5 below are the relations supported by ProtogaLattice. From a high-level, $\Xi_{f,n,B}$ is the base relation (i.e., the relation satisfied by fresh witnesses) and $\Xi_{f,n,B}^{\text{acc}}$ is the accumulator relation, both restricted to Ajtai commitments over modules.

Definition 5. *Let q be a prime, $\mathcal{R}_q$ be a cyclotomic ring, κ, n, m, B be integers. Let $\mathbf{A} \in \mathcal{R}_q^{\kappa \times m}$ be an Ajtai commitment key and $f : \mathcal{R}_q^m \rightarrow \mathcal{R}_q^n$ be a polynomial map such that $f(\mathbf{0}_m) = \mathbf{0}_n$. We define the input relation for the fresh witnesses as follows:*

$$\Xi_{f,n,B} := \left\{ (\mathbf{x} = \mathbf{t} \in \mathcal{R}_q^\kappa; \mathbf{w} \in \mathcal{R}_q^m) : \begin{array}{l} \mathbf{A}\mathbf{w} = \mathbf{t} \bmod q, \quad \|\mathbf{w}\| \leq B, \\ f(\mathbf{w}) = \mathbf{0}_n \end{array} \right\}$$

And the relation for the accumulator as follows:

$$\Xi_{f,n,B}^{\text{acc}} := \left\{ (\mathbf{x} = (\mathbf{t}, \boldsymbol{\beta}, e) \in \mathcal{R}_q^{\kappa + \log n + 1}; \mathbf{w} \in \mathcal{R}_q^m) : \begin{array}{l} \mathbf{A}\mathbf{w} = \mathbf{t} \bmod q, \\ \|\mathbf{w}\| \leq B \\ \sum_{i=1}^n \text{pow}_i(\boldsymbol{\beta}) f_i(\mathbf{w}) = e \end{array} \right\}.$$

where $\text{pow}_i(\boldsymbol{\beta})$ are as in Definition 4.

We omit the public parameters q, κ, m, p in the subscripts for the sake of readability.

While the relations $\Xi_{f,n,B}$ and $\Xi_{f,n,B}^{\text{acc}}$ are our target statements, the extractors of our folding and bootstrapping backbones only recover a *relaxed opening* to the input commitments. More precisely, for an input commitment-witness pair $(\mathbf{t}, \mathbf{w})$ in $\Xi_{f,n,B}$ or $\Xi_{f,n,B}^{\text{acc}}$, both extractors output a witness $\mathbf{w}$ satisfying $\mathbf{A}\mathbf{w} = \mathbf{t} \pmod{q}$, but $\mathbf{w}$ itself is of arbitrarily large norm. Rather, our extractor outputs a small factor $s \in \mathcal{R}_q$ with bounded operator norm such that $\|\mathbf{s}\mathbf{w}\| \leq B$. Definition 6 captures the notion of weak opening for both fresh instances and accumulators.

Expressing usual constraint systems. The notation $\mathbf{t} = \mathbf{A}\mathbf{w}$ and $f(\mathbf{w}) = \mathbf{0}_n$ expresses usual constraint systems such as R1CS and CCS without assuming any structure on them. As an example, we illustrate how R1CS translates to our relations. Let $\mathbf{w} = (\mathbf{x}, \mathbf{w}_{\text{in}}, 1) \in \mathbb{F}^m$ be a R1CS extended witness such that $\mathbf{A}\mathbf{w} \circ \mathbf{B}\mathbf{w} - \mathbf{C}\mathbf{w} = \mathbf{0}_n$ for matrices $\mathbf{A}, \mathbf{B}, \mathbf{C} \in \mathbb{F}^{n \times m}$. Then, we can express such a R1CS relation in our constraint system by simply denoting $\mathbf{t} = \text{cm}(\mathbf{w})$ and letting $f_i(\mathbf{w}) = \mathbf{a}_i^T \mathbf{w} \cdot \mathbf{b}_i^T \mathbf{w} - \mathbf{c}_i^T \mathbf{w}$ for each $i \in [n]$, which is a degree-2 polynomial map.

We remark that the notation $\mathbf{t} = \mathbf{A}\mathbf{w}$ is actually a simplification, as we can actually fold transcripts of (committed) special-sound protocol transcripts as described in Protostar [BC23]. For these, we can fold committed transcripts $(\mathbf{t}_0, c_1, \mathbf{t}_1, \dots, c_\mu, \mathbf{t}_\mu)$, where $\mathbf{t}_i = \mathbf{A}\mathbf{w}_i$ and $\mathbf{w}_i$ are the original prover messages. The advantage of these is that they *additionally* support complex Plonkish-like arithmetizations with lookups and range checks expressed in more rounds. We provide extensive details in the Supplementary Material A.2.

Supporting field-native relations. ProtogaLattice is native to constraints expressed directly over the ring $\mathcal{R}_q$, as the polynomial map f is defined over $\mathcal{R}_q$. If the constraints are natively expressed over the field $\mathbb{F}_q$ instead, one needs to *embed* $\mathbb{F}_q$ constraints and witnesses into an equivalent formulation over rings. The naïve approach is to embed each $\mathbb{F}_q$ element as a single $\mathcal{R}_q$ element. This is valid but somewhat wasteful ($\mathcal{R}_q$ elements are much larger than $\mathbb{F}_q$ elements), so Latticefold [BC25a] instead embeds vectors $\mathbf{a} \in \mathbb{F}_q^d$ into a single $\mathcal{R}_q$ element using the slots of the NTT transform. ProtogaLattice is fully compatible with this approach. Other lattice-based folding schemes, including SuperNeo [NS26] and Cyclo [GLLO26], use a clever embedding technique that enables them to run the sumcheck prover over the field instead of over the ring. ProtogaLattice can be adapted to support this embedding, at the cost of an additional reduction of knowledge as done in Cyclo [GLLO26, Section 2.6]. However, since ProtogaLattice does not run the sumcheck protocol, whether such an approach is beneficial for ProtogaLattice is unclear.

4.2 Architecture of ProtogaLattice

The architecture of our folding scheme (Figure 1) requires the fresh witnesses to go through a *folding range proof* (Definition 7) to allow the extractor to recover the original witnesses from the extracted relaxed openings of the folding and bootstrapping modules.

Informally, folding range proofs (Definition 7) relax the norm constraint $\|\mathbf{w}\| \leq B$ to a norm constraint on a small multiple of the witness $\|\mathbf{s}\mathbf{w}\| \leq B$ at the cost of extra polynomial constraints. Notably, we require that there exists an extractor that finds the original $\mathbf{w}$ satisfying the tight norm constraint $\|\mathbf{w}\| \leq B$ given relaxed openings $(\mathbf{w}, s)$ for an $\|s\|_{\text{op}} \leq \varrho$ where $\varrho > 1$.

The output relation of folding range proofs may include extra information in the instance, which we attach as $\mathbf{x}_{\text{RP}}$ to the instance of our relations $\Xi_{f,n,B,\varrho}^{\text{fold}}$ and $\Xi_{f,n,B,\varrho}^{\text{acc}}$.

Definition 6. We use the notations of Definition 5 and let $n_{\text{RP}}, m_{\text{RP}}$ be integers, $\varrho \geq 1$ and $f_{\text{RP}} : \mathcal{R}_q^{m_{\text{RP}}} \mapsto \mathcal{R}_q^{n_{\text{RP}}}$ be a polynomial map. We define the two following relaxed relations for fresh witnesses $(\Xi_{f,n,B,\varrho}^{\text{fold}})$ and for accumulators $(\Xi_{f,n,B,\varrho}^{\text{acc}})$:

$$\begin{aligned} \Xi_{f,n,B,\varrho}^{\text{fold}} &:= \left\{ \begin{array}{l} \left(\mathbf{x} = (\mathbf{t}, \mathbf{x}_{\text{RP}}); (\mathbf{w}, s) \right) : \\ \mathbf{t} \in \mathcal{R}_q^\kappa, \mathbf{x}_{\text{RP}} \in \mathcal{R}_q^{m_{\text{RP}}}, \\ \mathbf{w} \in \mathcal{R}_q^m, s \in \mathcal{R}_q, \\ \|s\|_{\text{op}} \leq \varrho, \\ \mathbf{A}\mathbf{w} = \mathbf{t} \bmod q, \\ \|s\mathbf{w}\| \leq B, \\ f(\mathbf{x}_{\text{RP}}, \mathbf{w}) = \mathbf{0}_n \end{array} \right\}, & \Xi_{f,n,B,\varrho}^{\text{acc}} &:= \left\{ \begin{array}{l} \left(\mathbf{x} = (\mathbf{t}, \mathbf{x}_{\text{RP}}, \beta, e); (\mathbf{w}, s) \right) : \\ \mathbf{t} \in \mathcal{R}_q^\kappa, \mathbf{x}_{\text{RP}} \in \mathcal{R}_q^{m_{\text{RP}}}, \\ \beta \in \mathcal{R}_q^t, e \in \mathcal{R}_q, \\ \mathbf{w} \in \mathcal{R}_q^m, s \in \mathcal{R}_q, \\ \|s\|_{\text{op}} \leq \varrho, \\ \mathbf{A}\mathbf{w} = \mathbf{t} \bmod q, \quad \|s\mathbf{w}\| \leq B, \\ \sum_{i=1}^n \text{pow}_i(\beta) f_i(\mathbf{x}_{\text{RP}}, \mathbf{w}) = e \end{array} \right\}. \end{aligned}$$

We omit the following public parameters $q, \kappa, m, m_{\text{RP}}, p$ in the subscript for the sake of readability.

4.3 Folding Range Proof

Definition 7. We use the notations of Definitions 5 and 6 and let $n_{\text{RP}}, m_{\text{RP}}$ be integers and $f_{\text{RP}} : \mathcal{R}_q^{m_{\text{RP}}} \mapsto \mathcal{R}_q^{n_{\text{RP}}}$ be a polynomial map. An interactive protocol Π_{RP} is a folding range proof when Π_{RP} is a reduction of knowledge

$$\Xi_{f,n,B} \xrightarrow{\Pi_{\text{RP}}} \Xi_{(f,f_{\text{RP}}),n+n_{\text{RP}},B,1}^{\text{fold}}$$

in the completeness direction and

$$\Xi_{f,n,B} \xleftarrow{\Pi_{\text{RP}}} \Xi_{(f,f_{\text{RP}}),n+n_{\text{RP}},B,\varrho}^{\text{fold}}$$

in the knowledge soundness direction.

Definition 8. We use the notations of Definitions 5 to 7 and let $k, B_{\text{ext}}, \gamma, \gamma'$ be integers. An interactive protocol $\Pi_{\text{rel_fold}}$ is a relaxed folding scheme when $\Pi_{\text{rel_fold}}$ is a reduction of knowledge

$$(\Xi_{f,n,B,1}^{\text{fold}})^k \times \Xi_{f,n,\gamma,1}^{\text{acc}} \xrightarrow{\Pi_{\text{rel_fold}}} \Xi_{f,n,\gamma',1}^{\text{acc}}$$

in the completeness direction and

$$(\Xi_{f,n,B_{\text{ext}},\varrho}^{\text{fold}})^k \times \Xi_{f,n,\gamma'+kB_{\text{ext}},\varrho}^{\text{acc}} \xleftarrow{\Pi_{\text{rel_fold}}} \Xi_{f,n,\gamma',1}^{\text{acc}}$$

in the soundness direction.

Composing Π_{RP} with $\Pi_{\text{rel_fold}}$ recovers the fresh witnesses tightly, thanks to the folding range proof. The accumulator, however, needs no range proof in ProtogaLattice's architecture. This is because the accumulator is not extracted but *reconstructed*. We can do this thanks to the application

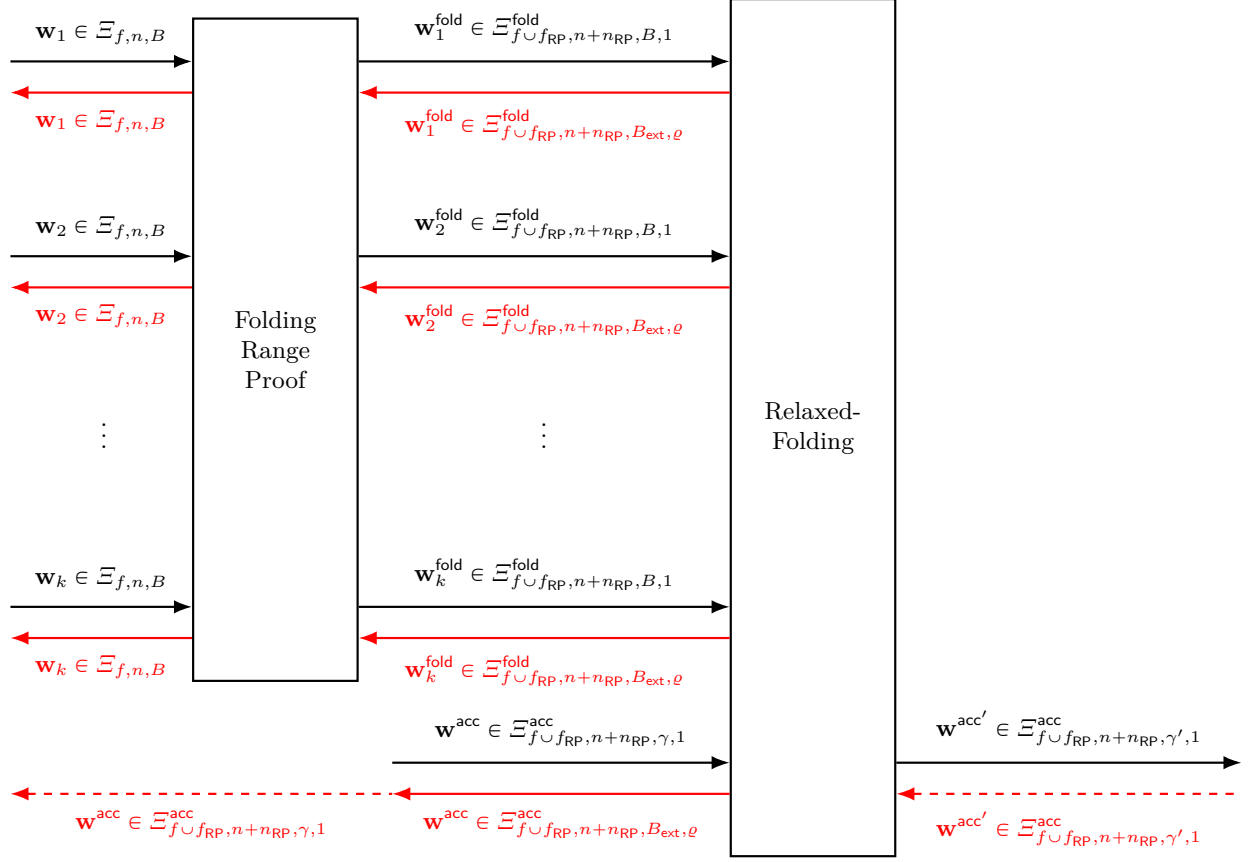


Fig. 1. Folding scheme from Folding Range Proof + Relaxed-Folding architecture. Black arrows represent protocol inputs and protocol outputs in the completeness direction. Red arrows represent protocol inputs and outputs in the soundness (extraction) direction. Dotted red arrows represent the extraction of the accumulator that ProtogaLattice can carry out without additional explicit range checks. Notations follow Definitions 5 to 8.

of Cyclo’s trick [GLLO26], i.e., combining the accumulator with a coefficient 1. Then, subtracting the (tight) fresh witnesses from the output accumulator yields it directly.

In the following result, we show that for folding schemes applying Cyclo’s trick, such as ProtogaLattice, the construction Π_{fold} in Figure 1 is a tight folding scheme.

Corollary 1. *We use the notations from Definitions 5 to 8. Let Π_{RP} be a folding range proof and $\Pi_{\text{rel_fold}}$ a relaxed folding scheme such that the output accumulator is computed as*

$$\mathbf{w}^* = \mathbf{w}_0 + \sum_{j \in [k]} c_j \mathbf{w}_j$$

for some coefficients c_j determined by the transcript. Then Π_{fold} of Figure 1 is a folding scheme

$$(\Xi_{f,n,B})^k \times \Xi_{f \cup f_{\text{RP}}, n+n_{\text{RP}}, \gamma_{\text{acc}}, 1}^{\text{acc}} \xrightarrow{\Pi_{\text{fold}}} \Xi_{f \cup f_{\text{RP}}, n+n_{\text{RP}}, \gamma', 1}^{\text{acc}},$$

where $\gamma_{\text{acc}} = \gamma' + 2kTB$.

Proof (Sketch). The proof is fairly straightforward, so we omit the full details. The goal is to show that the extractor returns tight witnesses for all $k+1$ inputs, including the accumulator. Essentially, the extractor of the folding scheme runs the extractor of $\Pi_{\text{rel_fold}}$ to obtain relaxed openings for the k fresh inputs, and the extractor of Π_{RP} to remove their slack. Combined, they yield witnesses $\mathbf{w}_j$ with $(\mathbf{t}_j; \mathbf{w}_j) \in \Xi_{f,n,B}$ for every $j \in [k]$. For the accumulator, setting $\mathbf{w}_0 := \mathbf{w}^* - \sum_{j \in [k]} c_j \mathbf{w}_j$ gives us that, by linearity, $\mathbf{A}\mathbf{w}_0 = \mathbf{t}_0$, where $\|\mathbf{w}_0\| \leq \gamma' + 2kTB$. The fact that the accumulator is valid follows from a similar argument to Section D.4. We remark that no rewinding is needed in this step.

5 ProtogaLattice

5.1 Overview of Our Techniques

We want to build an algebraic folding scheme from lattices for the two relations $\Xi_{f,n,\gamma,1}^{\text{acc}} \times (\Xi_{f,n,B,1}^{\text{fold}})^k \rightarrow \Xi_{f,n,\gamma',1}^{\text{acc}}$ defined in Definition 5 where the accumulator $\mathbf{w}^*$ is defined as

$$\mathbf{w}^*(X) = \sum_{i=0}^k L_i(X) \mathbf{w}_i.$$

We begin by investigating the simpler case $k = 1$. When folding one fresh witness $\mathbf{w}_1$ with an accumulator $\mathbf{w}_0^*$, we end up with an accumulator $\mathbf{w}_1^*$ as follows:

$$\mathbf{w}_1^* = L(X) \mathbf{w}_0^* + (1 - L(X)) \mathbf{w}_1.$$

In this simple case, we can choose $\mathbb{H} = \{0, 1\}$, then $L(X) = X$. In other words, for a small challenge $\gamma \in \mathcal{C}$, the linear combination coefficients are γ and $1 - \gamma$, which are both small. With $\mathcal{C} \subset \mathcal{R}_q$ a strong sampling set of small elements, T the expansion factor of $\mathcal{C}$, and B the norm bound for both $\mathbf{w}_0^*$ and $\mathbf{w}_1$, we obtain

$$\|\mathbf{w}_1^*\| \leq (2T + 1)B.$$

However, this approach does not scale for more than $k = 1$, as the Lagrange polynomials are of degree k , blowing up the norm. Hence, if we have k fresh witnesses $\mathbf{w}_1, \dots, \mathbf{w}_k$ and one accumulator $\mathbf{w}_0^*$, we can proceed recursively and fold each of the $\mathbf{w}_j$'s with the previous accumulator. Specifically, for all $j \in [k]$, we have the following recursive accumulator

$$\mathbf{w}_j^* := X_j \mathbf{w}_{j-1}^* + (1 - X_j) \mathbf{w}_j,$$

where X_j is the indeterminate for the evaluation of the degree-1 Lagrange polynomials at the challenge γ_j of the j -th folding step.

Our goal, however, is to support folding multiple witnesses at once. For this, we explicitly study the sequence $(\mathbf{w}_j^*)_{j \in [0,k]}$ in Lemma 4, which gives us the closed formula

$$\mathbf{w}_k^* = \left(\prod_{j=1}^k X_j \right) \mathbf{w}_0 + \sum_{j=1}^k \left((1 - X_j) \prod_{\ell=j+1}^k X_\ell \right) \mathbf{w}_j.$$

We notice that if all the challenges $(\gamma_j)_{j \in [k]}$ are drawn independently from $\mathcal{C}$, the coefficient c_0 of the first witness $\mathbf{w}_0$ (since it is the first one, we have $\mathbf{w}_0 = \mathbf{w}_0^*$) is the product of all γ_j 's,

thus c_0 has coefficients of size $O(N^{k-1})$. Fortunately, Lemma 4 also provides a change of variables $Y_j := \prod_{\ell=j+1}^k X_\ell$, which simplifies the closed form of $\mathbf{w}_k^*$ to the following

$$\mathbf{w}_k^* = Y_0 \mathbf{w}_0 + \sum_{j=1}^k (Y_j - Y_{j-1}) \mathbf{w}_j.$$

Next, we wish to turn the above relation into a folding scheme for k witnesses and an accumulator, with a single-round challenge $(y_0, \dots, y_{k-1}) \leftarrow \mathcal{C}^k$.

Multivariate polynomials for folding over rings. A crucial fact needed to pack all the folding iterations is the following: the first two rounds (with the random challenges α, δ) only need to be completed once at the beginning of the k iterations, provided that the commitments to all the witnesses $\mathbf{w}_0, \dots, \mathbf{w}_k$ are sent prior to α and δ .

The last major challenge is to establish a generalization for the core relation over Lagrange polynomials Lemma 3 over the rings. We provide such a generalization in Theorem 1. Recall that intuitively, Lemma 3 claims that the composition of a polynomial map f and the linear combination defined by the Lagrange polynomials behaves as a linear function within the interpolation domain (i.e. modulo the vanishing polynomial). In the univariate case, the vanishing ideal of $\mathbb{H}$ is the principal ideal $\langle Z \rangle$, but in our multivariate case, the vanishing ideal $\mathcal{I}$ (see Definition 9) that we consider has instead $2k - 1$ generators. After a careful analysis of $\mathcal{I}$, we establish a reduced Gröbner basis $\mathcal{G}$ (generalized to rings, see Section D.1) in Proposition 1. From a very high-level, a reduced Gröbner basis (even over a ring) allows to perform Euclidean division, with guarantees on uniqueness of the remainder and quotient. The latter properties are enough to extend the univariate Protogalaxy to our multivariate setting.

5.2 Protogalaxy's Algebraic Folding PGL-Fold

We present the main folding protocol in Figure 2. We start by introducing a technical lemma that finds a closed-form expression for accumulators.

Lemma 4. *Consider $k + 1$ vectors $\mathbf{w}_0, \dots, \mathbf{w}_k \in \mathcal{R}_q^m$ and k variables $X_1, \dots, X_k$. For all $j \in [k]$, define*

$$\mathbf{w}_j^* = X_j \mathbf{w}_{j-1}^* + (1 - X_j) \mathbf{w}_j.$$

We have the following closed form for $\mathbf{w}_k^$:*

$$\mathbf{w}_k^* = \left(\prod_{j=1}^k X_j \right) \mathbf{w}_0 + \sum_{j=1}^k \left((1 - X_j) \prod_{\ell=j+1}^k X_\ell \right) \mathbf{w}_j.$$

Equivalently, with respect to the variables $Y_j := \prod_{\ell=j+1}^k X_\ell$ for all $j \in [0, k]$, we have

$$\mathbf{w}_k^* = Y_0 \mathbf{w}_0 + \sum_{j=1}^k (Y_j - Y_{j-1}) \mathbf{w}_j.$$

Notice that $Y_k = \prod_{\ell=k+1}^k X_\ell = 1$.

Proof. We define the sequence $(\mathbf{w}_j^*)_{j \in [k]}$ as follows:

1. $\mathbf{w}_0^* = \mathbf{w}_0$
2. For all $j \in [k]$ $\mathbf{w}_j^* = X_j \mathbf{w}_{j-1}^* + (1 - X_j) \mathbf{w}_j$.

We prove by induction on k that:

$$\mathbf{w}_k^* = \left(\prod_{j=1}^k X_j \right) \mathbf{w}_0 + \sum_{j=1}^k \left((1 - X_j) \prod_{\ell=j+1}^k X_\ell \right) \mathbf{w}_j.$$

Initialization. For $k = 1$, we have $\mathbf{w}_1^* = X_1 \mathbf{w}_0^* + (1 - X_1) \mathbf{w}_1$. The formula for the closed form of $\mathbf{w}_1^*$ gives:

$$\begin{aligned} \mathbf{w}_1^* &= \left(\prod_{j=1}^1 X_j \right) \mathbf{w}_0 + \sum_{j=1}^1 \left((1 - X_j) \prod_{\ell=j+1}^1 X_\ell \right) \mathbf{w}_j \\ &= X_1 \mathbf{w}_0 + (1 - X_1) \cdot 1 \cdot \mathbf{w}_1, \end{aligned}$$

thus the formula holds for $k = 1$.

Induction. We have $\mathbf{w}_{k+1}^* = X_{k+1} \mathbf{w}_k^* + (1 - X_{k+1}) \mathbf{w}_{k+1}$. We substitute $\mathbf{w}_k^*$ and obtain:

$$\begin{aligned} \mathbf{w}_{k+1}^* &= X_{k+1} \left[\left(\prod_{j=1}^k X_j \right) \mathbf{w}_0 + \sum_{j=1}^k \left((1 - X_j) \prod_{\ell=j+1}^k X_\ell \right) \mathbf{w}_j \right] + (1 - X_{k+1}) \mathbf{w}_{k+1} \\ &= \left(\prod_{j=1}^{k+1} X_j \right) \mathbf{w}_0 + \sum_{j=1}^k \left((1 - X_j) \prod_{\ell=j+1}^{k+1} X_\ell \right) \mathbf{w}_j + (1 - X_{k+1}) \prod_{\ell=k+2}^{k+1} X_\ell \mathbf{w}_{k+1} \\ &= \left(\prod_{j=1}^{k+1} X_j \right) \mathbf{w}_0 + \sum_{j=1}^{k+1} \left((1 - X_j) \prod_{\ell=j+1}^{k+1} X_\ell \right) \mathbf{w}_j, \end{aligned}$$

which completes the proof. □

Definition 9. Let $\mathcal{Y} = (Y_0, \dots, Y_{k-1})$. For all $i, j \in [0, k-1]$, we define the polynomial

$$Z_{ij}(\mathcal{Y}) := Y_i(Y_j - 1).$$

Furthermore, we define the ideal $\mathcal{I} \subset \mathcal{R}_q[\mathcal{Y}]$ as

$$\mathcal{I} = \langle Z_{0,0}, \dots, Z_{k-1,k-1}, Z_{1,0}, \dots, Z_{k-1,k-2} \rangle$$

Theorem 1. Let $f : \mathcal{R}_q^m \rightarrow \mathcal{R}_q$ be a polynomial map of degree d and $w_0, \dots, w_k \in \mathcal{R}_q^m$. We furthermore assume that the Y_j 's and the $1 - Y_j$'s are either invertible elements in $\mathcal{R}_q$ or 0. Then the following holds:

$$f \left(Y_0 w_0 + \sum_{j=1}^k (Y_j - Y_{j-1}) w_j \right) - \left(Y_0 f(w_0) + \sum_{j=1}^k (Y_j - Y_{j-1}) f(w_j) \right) \in \mathcal{I}.$$

For better readability, we relegate the proof to Appendix D.2.

Proposition 1 (Gröbner basis of $\mathcal{I}$). Recall that $Z_{i,j}(\mathcal{Y}) = Y_i Y_j - Y_i$ and $\mathcal{M} = \mathcal{R}_q[\mathcal{Y}]$. Let $\mathcal{G} := \{Z_{i,j}, 0 \leq j \leq i \leq k-1\}$ and

$$\mathcal{S}(\mathcal{G}) = \{(Q_{ij})_{\substack{i,j \in [0,k-1] \\ j \leq i}} \in (\mathcal{R}_q[Y_0, \dots, Y_{k-1}])^{k(k+1)/2}, \sum_{\substack{i,j \in [0,k-1] \\ j \leq i}} Q_{ij} Z_{ij} = 0\}$$

be the set of syzygies of $\mathcal{G}$. Let $\mathcal{R}$ be a set of coset representatives of $\mathcal{M}/\mathcal{I}$ and $\mathcal{S}$ be a set of coset representatives of the set of $\mathcal{M}^{k(k+1)/2}/\mathcal{S}(\mathcal{G})$.

The set $\mathcal{G}$ is a reduced Gröbner basis of $\mathcal{I}$. More specifically, for all $P \in \mathcal{M}$, there exists a unique remainder $R \in \mathcal{R}$ and a unique vector of quotients $(Q_{ij})_{\substack{i,j \in [0,k-1] \\ j \leq i}} \in \mathcal{S}$ such that

$$P = \sum_{\substack{i,j \in [0,k-1] \\ j \leq i}} Q_{ij} Z_{ij} + R.$$

For the sake of clarity, we may omit the heavy subscript and write $(Z_{ij})_{ij}$ for $(Z_{ij})_{\substack{i,j \in [0,k-1] \\ j \leq i}}$ when it is clear from context.

Note that $\mathcal{G} \subseteq \mathcal{I}$, since the relations $Y_{i+1}Y_i = Y_{i+1}$ give $Y_i Y_j = Y_i$ for every $j \leq i$ by induction. Hence, every Z_{ij} with $j \leq i$ lies in $\mathcal{I}$. Thus $\mathcal{G}$ generates $\mathcal{I}$ even though it is strictly larger than the generating set of Definition 9.

On the practical cost of computing a reduction modulo $\mathcal{I}$. The reduced Gröbner basis explicitly given in Proposition 1 has a very rich structure which enables significant implementation speedups. More precisely, since $\mathcal{G}$ contains Z_{ii} for all $i \in [0, k-1]$, we have $Y_i^2 = Y_i$ thus $Y_i^\alpha = Y_i$ for any exponent $\alpha \geq 1$, i.e all powers are reduced to 1. Additionally, $\mathcal{G}$ contains Z_{ij} for all $j \leq i$, thus any product of two variables $Y_i Y_j$ is reduced to its variable of highest index $Y_{\max\{i,j\}}$. Combining both remarks, we obtain that for a monomial of the form $M = Y_0^{\alpha_0} \dots Y_{k-1}^{\alpha_{k-1}}$, M 's remainder modulo $\mathcal{G}$ is simply $Y_{\max\{i \text{ st } \alpha_i \neq 0\}}$. Finally, we note that these polynomials are *small* as their size is constant on m or n , so this reduction does not dominate in the overall prover time.

Linear norm growth for the accumulator. Iteratively folding by applying a linear combination with random weights to the witnesses to be fold results in a multiplicative norm growth for the output accumulator $\mathbf{w}^*$ in the number of folding steps. At each iteration, the prior accumulator is multiplied by a constant coefficient, carrying the increasing norm further. Specifically, at iteration ℓ , $\mathbf{w}_\ell^*$ is constructed as a linear combination in which one of the terms is $c_0 \mathbf{w}_{\ell-1}^*$, therefore, at each new folding step, the norm of the accumulator is multiplied by a constant factor that is the operator norm of c_0 .

First step is to apply the key technique to reduce norm growth introduced by Cyclo [GLLO26] in sum-check based folding schemes. The idea behind Cyclo's trick is that the extractor only needs k independent challenges to recover $k+1$ (relaxed) input witnesses. Since one of the challenges can be fixed, we fix the coefficient c_0 to 1, thus reducing the norm growth from exponential to linear in the recursive depth. Unfortunately, the trick does not apply straightforwardly to us, as the coefficients of our $k+1$ witnesses in the output accumulator are already determined by only k

independent challenges. This is due to the partition of unity property of the Lagrange polynomials (that is, they always sum to 1).

Our solution is to artificially increase by 1 the number of independent challenges, allowing us to fix the accumulator's coefficient to 1. In our supported relations, in line with most arithmetizations (e.g R1CS or CCS), the constraints polynomial f is homogenous, i.e. $f(\mathbf{0}) = 0$. Since the zero witness is in the input relation and its commitment is also a zero vector, it follows that we can fold a "ghost" zero witness without changing the protocol. Rather, the only difference is that a new variable Y_k is appended to the list of challenges, therefore the last coefficient is $L_k(\mathcal{Y}) = Y_k - Y_{k-1}$ instead of $1 - Y_{k-1}$, which is exactly what is needed.⁸

Definition 10 (Multivariate Lagrange polynomials for Cyclo's trick). *Let k be a positive integer. We define our multivariate Lagrange polynomials over $\mathcal{R}[Y_0, \dots, Y_k]$ as $L_0(\mathcal{Y}) = Y_0$, $L_{k+1}(\mathcal{Y}) = 1 - Y_k$ and for all $j \in [k]$, $L_j(\mathcal{Y}) = Y_j - Y_{j-1}$.*

The proof of Proposition 1 can be found in Appendix D.3. We now state the main theorem and refer the reader to the Appendix D.4 for the proof.

Theorem 2. *Let λ be a security parameter, $\mathcal{C} \subset \mathcal{R}_q$ be a strong sampling set, B be a norm bound for the fresh input witnesses, γ be a norm bound for the accumulator and $\gamma' = \gamma + (2kT + 1)B$.*

*Under the hardness of $\text{MSIS}_{\kappa', m, 2(2T)^{2k}(2\gamma' + B)}^{\infty, q}$, **PGL-Fold** (Figure 2) is a reduction of knowledge*

$$\begin{array}{ccc} (\Xi_{f,n,B,1}^{\text{fold}})^k \times \Xi_{f,n,\gamma,1}^{\text{acc}} & \longrightarrow & \Xi_{f,n,\gamma',1}^{\text{acc}} \\ (\Xi_{f,n,B_{\text{ext},\varrho}}^{\text{fold}})^k \times \Xi_{f,n,B_{\text{ext},\varrho}}^{\text{acc}} & \longleftarrow & \Xi_{f,n,\gamma',1}^{\text{acc}} \end{array}$$

5.3 A Witness Bootstrapping Protocol

Witness decomposition for non-linear relations. Despite the integration of Cyclo's trick to PGL-Fold, the output accumulator $\mathbf{w}^*$'s norm is still linearly growing with each folding step. This restricts the recursive composition to a bounded folding depth. We overcome this, with a new norm bootstrapping technique unlocking unbounded folding composition for any IVC application.

We propose a so-called *witness bootstrapping* protocol, which is a reduction of knowledge

$$\begin{array}{ccc} \Xi_{f,n,\gamma',1}^{\text{acc}} & \longrightarrow & \Xi_{f,n,\gamma,1}^{\text{acc}} \\ \Xi_{f,n,\gamma',\varrho}^{\text{acc}} & \longleftarrow & \Xi_{f,n,\gamma,1}^{\text{acc}} \end{array}$$

for $\gamma \leq \gamma'$. In other words, our witness bootstrapping protocol takes an accumulator of norm γ' and returns another accumulator with a lower norm γ , thus allowing the folding to continue.

The witness bootstrapping protocol plays a similar role as the witness decompositions in LatticeFold(+) and (Super)Neo. In those schemes, a witness $\mathbf{w}$ with norm bound γ' is decomposed in k' witnesses in basis b as follows

$$\mathbf{w} = \sum_{j=0}^{k'-1} b^j \mathbf{w}_j,$$

⁸ Additionally, there is a polynomial $L_{k+1}(\mathcal{Y}) = 1 - Y_k$, though it is multiplied by a zero-witness, so it does not actually contribute to the accumulator.

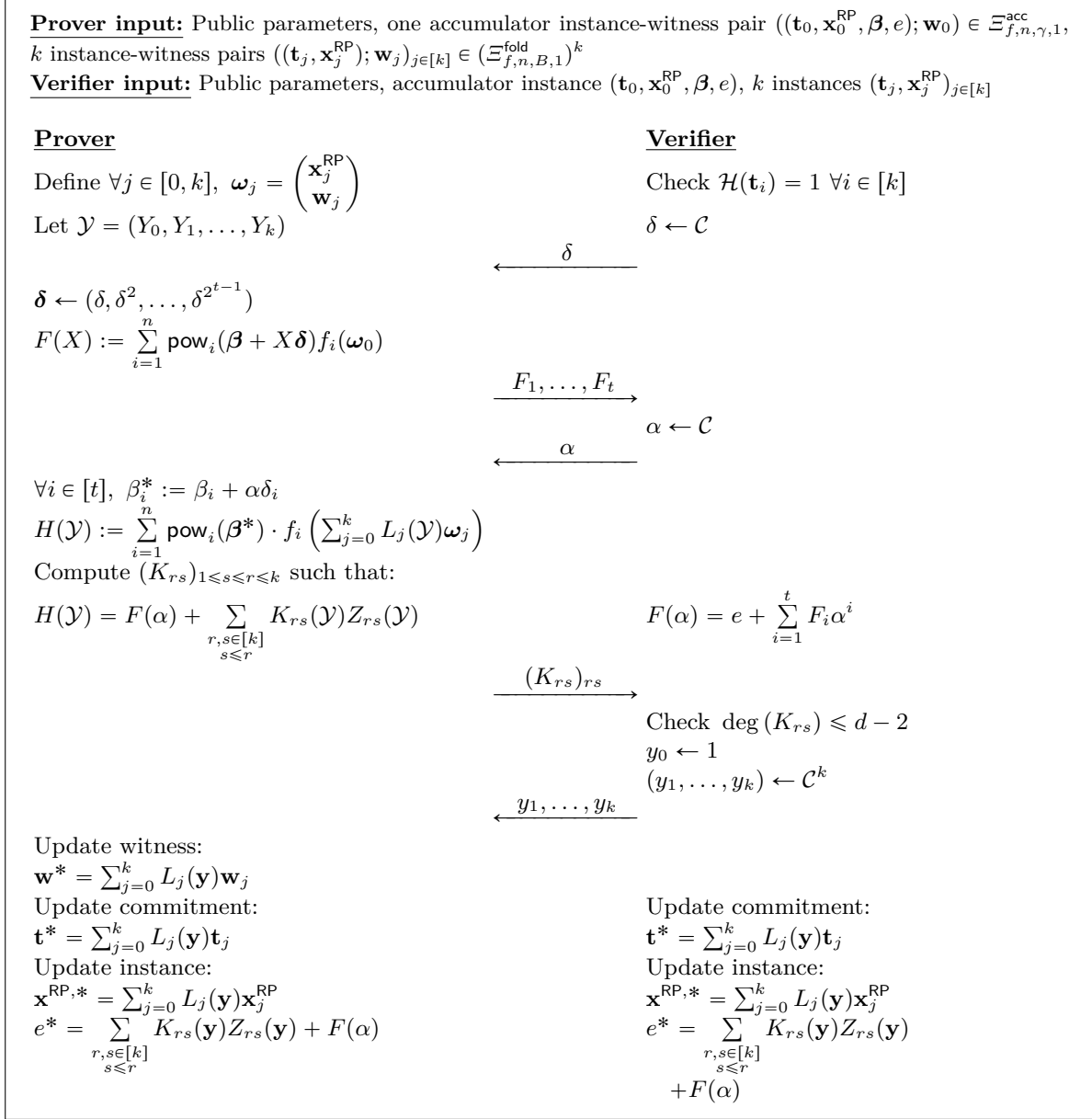


Fig. 2. Folding protocol PGL-Fold: $\Xi_{f,n,\gamma,1}^{\text{acc}} \times (\Xi_{f,n,B,1}^{\text{fold}})^k \rightarrow \Xi_{f,n,\gamma',1}^{\text{acc}}$. The check $\text{Check } \mathcal{H}(\mathbf{t}_i) = 1 \ \forall i \in [k]$ is only necessary if the relation is given by a special-sound protocol with at least 1 round; see Section A.2 for details.

where $b^{k'} \geq \gamma' - 1$. Since the relations considered by those schemes are linear, connecting the relations satisfied by the $\mathbf{w}_j$'s and that of $\mathbf{w}$ is straightforward. In our case, for an accumulator instance-witness pair $((\mathbf{t}, \beta, e), \mathbf{w}) \in \Xi_{f,n,\gamma',1}^{\text{acc}}$, the relation is not linear, thus connecting the e_j 's defined as follows

$$\forall j \in [k'], e_j := \sum_{i=1}^n \text{pow}_i(\beta) f_i(\mathbf{w}_j)$$

to e is not trivial. Our idea is to rely Theorem 1, our generalization of Lemma 3 to multivariate polynomials.

For clarity, we explain the intuition in the univariate case. Let us assume that there exists an evaluation point D such that for all $j \in [k']$, we have $L_j(D) = b^{j-1}$. It follows that under Lemma 3, we have

$$\sum_{j=1}^{k'} b^{j-1} e_j = e + Z(D)K(D), \quad (3)$$

for a quotient K . Summing up, the protocol that we obtain so far looks as follows: first, the prover decomposes $\mathbf{w}$ in base b and obtains $\mathbf{w}_1, \dots, \mathbf{w}_{k'}$. Then, he computes the commitments $\mathbf{t}_j = \mathbf{A}\mathbf{w}_j$, and the errors e_j such that for all $j \in [k']$, $((\mathbf{t}_j, \beta, e_j); \mathbf{w}_j) \in \Xi_{f,n,b,1}^{\text{acc}}$. Next, the prover computes the polynomial

$$H(X) := \sum_{i=1}^n \text{pow}_i(\beta) f_i \left(\sum_{j=1}^{k'} L_j(X) \mathbf{w}_j \right),$$

and the quotient K of H in the division by Z . The prover sends K to the verifier, who checks that Equation (3) holds. There are at least two problems with the aforementioned construction:

1. The prover needs to convince the verifier that K is correctly formed,
2. For a given b , there may not exist a tuple D satisfying for all $j \in [k']$ $L_j(D) = b^{j-1}$.

We solve the first problem by actually finishing the folding of the $\mathbf{w}_j$'s: after receiving K , the verifier sends a challenge γ and thus creates a witness $\mathbf{w}^* := \sum_{j=0}^{k'-1} L_j(\gamma) \mathbf{w}_j$ that satisfies

$$\sum_{i=1}^n \text{pow}_i(\beta) f_i(\mathbf{w}^*) = \sum_{j=0}^{k'-1} L_j(\gamma) e_j =: e^*.$$

After a careful analysis, we find that the well-formedness of K is ensured by the soundness of the folding of the $\mathbf{w}_j$'s into $\mathbf{w}^*$.

The second problem comes from the fact that the sum of the Lagrange polynomials is constant equal to 1, therefore when $\sum_{j=1}^{k'} b^{j-1} \neq 1$ (this condition is always satisfied for practical values of b), no such D exists. Similarly to PGL-Fold, the bootstrapping protocol PGL-Boot also folds a ghost zero-witness that artificially increases the number of independent challenges. This enables us to get an extra degree of freedom, and we can set D as follows:

$$D_j = \sum_{i=0}^j b^i = (b^{j+1} - 1)/(b - 1)$$

After careful analysis in Theorem 3, the intuitive overview presented in the univariate case extends to the multivariate case, and yields the protocol PGL-Boot presented in Figure 3.

Prover input: Public parameters, one accumulator instance-witness pair $((\mathbf{t}_0, \mathbf{x}_0^{\text{RP}}, \beta, e); \mathbf{w}_0) \in \Xi_{f,n,\gamma',1}^{\text{acc}}$.

Verifier input: Public parameters, accumulator instance $(\mathbf{t}, \mathbf{x}_0^{\text{RP}}, \beta, e)$.

Prover

Compute base b decomposition $\sum_{j=0}^{k'-1} \mathbf{w}_j b^j$
 Compute base b decomposition $\sum_{j=0}^{k'-1} \mathbf{x}_j^{\text{RP}} b^j$
 Let $\forall j \in [0, k' - 1], \omega_j = \begin{pmatrix} \mathbf{x}_j^{\text{RP}} \\ \mathbf{w}_j \end{pmatrix}$
 Define $\mathcal{Y} := (Y_0, Y_1, \dots, Y_{k'-1})$
 Commit $\forall j \in [0, k' - 1], \mathbf{t}_j = \mathbf{A} \mathbf{w}_j$
 $\forall j \in [0, k' - 1], e_j = \sum_{i \in [n]} \text{pow}_i(\beta) f_i(\omega_j)$
 $H(\mathcal{Y}) := \sum_{i=1}^n \text{pow}_i(\beta) f_i \left(\sum_{j=0}^{k'-1} L_j(\mathcal{Y}) \omega_j \right)$
 Compute $(K_{rs})_{0 \leq s \leq r \leq k'-1}$ such that:
 $H(\mathcal{Y}) = \sum_{j=0}^{k'-1} L_j(\mathcal{Y}) e_j + \sum_{r,s} Z_{rs}(\mathcal{Y}) K_{rs}(\mathcal{Y})$

$$(\mathbf{t}_j, e_j)_{j \in [0, k'-1]} \xrightarrow{(K_{rs})_{rs}}$$

Verifier

Check $\deg(K_{rs}) \leq d - 2$
 Check $\sum_{j=0}^{k'-1} b^j \mathbf{t}_j \stackrel{?}{=} \mathbf{t}$
 Check $\sum_{j=0}^{k'-1} b^j e_j \stackrel{?}{=} e$
 $+ \sum_{r,s} Z_{rs}(D) K_{rs}(D)$
 $y_0 \leftarrow 1$
 $(y_1, \dots, y_{k'-1}) \leftarrow \mathcal{C}^{k'-1}$

$$\xleftarrow{y_1, \dots, y_{k'-1}}$$

Define for all $j \in [0, k' - 1]$
 $c_j = L_j(\mathbf{y})$

Define for all $j \in [0, k' - 1]$
 $c_j = L_j(\mathbf{y})$

Update witness:

$$\mathbf{w}^* = \sum_{j=0}^{k'-1} c_j \mathbf{w}_j$$

Update commitment:

$$\mathbf{t}^* = \sum_{j=0}^{k'-1} c_j \mathbf{t}_j$$

Update instance:

$$\mathbf{x}^{\text{RP},*} = \sum_{j=0}^{k'-1} L_j(\mathbf{y}) \mathbf{x}_j^{\text{RP}}$$

$$e^* = \sum_{r,s} K_{rs}(\mathbf{y}) Z_{rs}(\mathbf{y}) + \sum_{j=0}^{k'-1} c_j e_j$$

Update commitment:

$$\mathbf{t}^* = \sum_{j=0}^{k'-1} c_j \mathbf{t}_j$$

Update instance:

$$\mathbf{x}^{\text{RP},*} = \sum_{j=0}^{k'-1} L_j(\mathbf{y}) \mathbf{x}_j^{\text{RP}}$$

$$e^* = \sum_{r,s} K_{rs}(\mathbf{y}) Z_{rs}(\mathbf{y}) + \sum_{j=0}^{k'-1} c_j e_j$$

Fig. 3. Witness Bootstrapping protocol PGL-Boot: $\Xi_{f,n,\gamma',1}^{\text{acc}} \rightarrow \Xi_{f,n,\gamma,1}^{\text{acc}}$.

Theorem 3. Let $k' \geq 2$ an integer, $b = \lceil (\gamma' - 1)^{1/k'} \rceil$, $\gamma = (2T(k' - 1) + 1)b$ the target bound for the output accumulator, and $D \in \mathcal{R}_{q,q}^{k'-1}$ be the evaluation tuple such that $\forall j \in [0, k' - 1]$, $L_j(D) = b^j$.

Under the hardness of $\text{MSIS}_{\kappa', m, 2(k'-1)(2T)^{2k'}\gamma(2\gamma')}^{\infty, q}$, **PGL-Boot** defined in Figure 3 is a reduction of knowledge

$$\begin{array}{ccc} \Xi_{f,n,\gamma',1}^{\text{acc}} & \longrightarrow & \Xi_{f,n,\gamma,1}^{\text{acc}} \\ \Xi_{f,n,\gamma_{\text{ext}},(2T)^{k'}}^{\text{acc}} & \longleftarrow & \Xi_{f,n,\gamma,1}^{\text{acc}} \end{array}$$

where $\gamma_{\text{ext}} := (k' - 1)\gamma(2T)^{k'}(2\gamma')$.

6 Performance and Efficiency Estimates

In this section, we provide explicit formulae for the proof sizes of **PGL-Fold** and **PGL-Boot**, then we provide concrete parameter settings and efficiency estimates for **PGL-Fold**. We summarize the parameters involved in this paper in a glossary Table 3.

Notation	Description	Instantiation example
q	Prime modulus	$\log q \simeq 64$
N	Degree of the cyclotomic ring	32
d'	Number of the CRT fields	4
κ	Height of the commitment matrix for fresh inputs	54
κ'	Height of the commitment matrix for accumulator input	54
n	Number of ring constraints	$2^{24.4}$
t	$t := \lceil \log n \rceil$	25
m	Length of the witness (over $\mathcal{R}_q$)	$2^{24.4}$
d	Degree of f	2
k	Number of fresh witnesses	1
k'	Number of decomposition witnesses	—
L	Maximum number of folding steps without PGL-Boot	64
B	Fresh input witnesses norm bound	1 (ternary)
γ	Input accumulator norm bound	—
γ'	Output accumulator norm bound	—
b	Decomposition basis in PGL-Boot	—
B_C	Range of challenge coefficients	16
$\mathcal{C}$	Challenge set	$\{-7, \dots, 8\}^N$
T	Expansion factor of $\mathcal{C}$	480

Table 3. Glossary parameter table, showing notation, description and a concrete instantiation choice for **PGL-Fold**. We do not give concrete numbers for **PGL-Boot**-only parameters since we do not instantiate it explicitly.

6.1 Concrete sizes of **PGL-Fold** and **PGL-Boot**

All quantities are originally measured in arbitrary integers, arbitrary ring elements, or arbitrary challenge set elements. For each of these sets, we write $|\cdot|_{\text{bit}}$ the theoretical bit-size of elements of this set. We have

$$|\mathbb{Z}_q|_{\text{bit}} = \log q, \quad |\mathcal{R}_q|_{\text{bit}} = N \log q, \quad |\mathcal{C}|_{\text{bit}} = N \log_2(B_C + 1) = \lambda,$$

where $B_{\mathcal{C}} := 2^{\lambda/N} - 1$ is the smallest range of challenge coefficients such that the challenge set $\mathcal{C} = [-\lceil B_{\mathcal{C}}/2 \rceil + 1, \lceil B_{\mathcal{C}}/2 \rceil]^N$ satisfies $|\mathcal{C}| = 2^\lambda$. We note that the ℓ_∞ -operator-norm of challenges is upper bounded by $T := NB_{\mathcal{C}}$.

	PGL-Fold	PGL-Boot
Commitments	$(\kappa' + k \kappa) N \log q$	$k' \kappa' N \log q$
Instance	$(k + 1) \mathbf{x}^{\text{RP}} _{\text{bit}} + (t + 1) N \log q$	$k' \mathbf{x}^{\text{RP}} _{\text{bit}} + (t + 1) N \log q$
Prover msgs.	$\left(t + \frac{k(k+1)}{2} \binom{k+d-2}{k}\right) N \log q$	$\left(\frac{k'(k'-1)}{2} \binom{k'+d-3}{k'-1}\right) N \log q$
Challenges	$(k + 2) \mathcal{C} _{\text{bit}}$	$(k' - 1) \mathcal{C} _{\text{bit}}$
Rounds	3	1
MSIS bound	$(2T)^{2k} (k^2 + k + 1) (\gamma + (2kT + 1)B)$	$(2T)^{2k'} ((k')^2 + k' + 1) (\gamma + (2k'T + 1)b)$

Table 4. Theoretical sizes, in bits, of the two protocols. Here $t := \lceil \log n \rceil$, $d := \deg f$, k is the number of fresh instances folded per step and k' the number of decomposition witnesses.

6.2 Concrete Instantiation

We compute various sets of parameters for PGL-Fold together with the range proof described in Section C that reach $\lambda = 128$ computational bits of security, estimating the hardness of MSIS via the lattice-estimator library [APS15]. We target $L = 64$ folding steps.

We include the cost of the most efficient range proof available to our scheme, which we detail in Section C. Since this range proof is fit to prove that the input witnesses are ternary, i.e with infinity norm bound $B = 1$, we compensate to make a fair comparison by increasing the integer length mN to reach identical witness bit-length as other schemes. Concretely, our witness size of 0.108 GiB corresponds to Cyclo’s benchmark witness of integer dimension 2^{27} with norm bound 2^{10} . For concrete instantiations, we conservatively assume that $\kappa = \kappa'$, i.e that the commitment key that we use to commit to every fresh witness is the same as the commitment key used to commit to the final large accumulator.

The results presented in Table 5 show that the concrete communication of PGL-Fold is very efficient (under 10 KB) for degree-2 relations (e.g R1CS) and one fresh witness at a time ($k = 1$), and scales slightly faster than linearly with k . We note that part of the scaling with k is due to the constant number of folding steps $L = 64$, not number of fold-able witnesses. The effective number of foldable witnesses is given by kL , which hinders the efficiency of multi-folding. Another significant contribution to the total size as k grows is the exponential extraction slack in k due to the integration of Cyclo’s linear-growth trick. Notably, the communication cost scales with $\binom{k+d-2}{k}$, therefore high-degree relations (e.g $\deg(f) = 5$) are barely more expensive than degree-2 relations for $k = 1$, but scale very poorly as k grows.

On the Usage of AI Tools

We have not used generative AI to produce any of the contents in this paper, except for formatting of some tables and diagrams and for reviewing the consistency of some notation. The only exception is Section C, where we have used AI tools to import notation and equations from other papers.

$\deg(f)$	witness size(GiB)	k	$\log(q)$	κ	λ	$\log_2(B_{\text{MSIS}})$	Commit. (KB)	Inst. (KB)	Proof (KB)	Total (KB)
2	0.108	1	64	54	128.4	53.2	27.0	6.3	8.13	41.4
2	0.108	2	120	58	130.1	75.2	81.6	11.7	14.44	107.7
2	0.108	3	192	59	128.8	96.5	177.0	18.8	25.36	221.2
5	0.108	1	64	54	128.4	53.2	27.0	6.3	8.88	42.2
5	0.108	2	120	58	130.0	75.2	81.6	11.7	27.09	120.4
5	0.108	3	192	59	128.8	96.5	177.0	18.8	110.86	306.7
2	1.0	1	64	57	129.6	54.5	28.5	7.0	9.03	44.53
2	1.0	2	128	56	129.5	76.5	84.0	14.0	17.05	115.1
2	1.0	3	200	58	128.5	97.8	181.3	21.9	29.03	232.2
5	1.0	1	64	57	129.6	54.5	28.5	7.0	9.78	45.3
5	1.0	2	128	56	129.5	76.5	84.0	14.0	30.55	128.6
5	1.0	3	200	58	128.5	97.8	181.3	21.9	118.07	321.3

Table 5. Parameter selection and concrete sizes of PGL-Fold with the range proof described in Section C, targeting number of folding steps $L = 64$.

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A Deferred Preliminaries

A.1 Commitments

Definition 11. A commitment scheme is a tuple of algorithms $(\text{Setup}, \text{Com}, \text{VerCom})$ that work as follows:

- $\text{Setup}(1^\lambda) \rightarrow \text{ck}$ takes as input the security parameter in unary and returns the commitment key ck , and descriptions of the input space $\mathcal{D}$, commitment space $\mathcal{C}$ and opening space $\mathcal{O}$.
- $\text{Com}(\text{ck}, v) \rightarrow (c, o)$ takes the commitment key ck and a value $v \in \mathcal{D}$, and outputs a commitment c and an opening o .
- $\text{VerCom}(\text{ck}, c, v, o) \rightarrow b$ takes as input a commitment c , a value v and an opening o , and accepts ($b = 1$) or rejects ($b = 0$).

A commitment scheme should satisfy the correctness and binding properties, and often they also satisfies a hiding property.

Correctness. For all $\lambda \in \mathbb{N}$ and any input $v \in \mathcal{D}$ we have:

$$\Pr \left[\text{ck} \leftarrow \text{Setup}(1^\lambda), (c, o) \leftarrow \text{Com}(\text{ck}, v) : \text{VerCom}(\text{ck}, c, v, o) = 1 \right] = 1$$

Binding. For every polynomial-time adversary $\mathcal{A}$, the following probability is negligible in the security parameter

$$\Pr \left[\begin{array}{l} v \neq v' \wedge \text{VerCom}(\text{ck}, c, v, o) = 1 \\ \text{ck} \leftarrow \text{Setup}(1^\lambda), (c, v, o, v', o') \leftarrow \mathcal{A}(\text{ck}) ; \text{VerCom}(\text{ck}, c, v', o') = 1 \end{array} \right]$$

Hiding. For $\text{ck} \leftarrow \text{Setup}(1^\lambda)$ and $\forall v, v' \in \mathcal{D}$, the following two distributions are statistically close: $\text{Com}(\text{ck}, v) \approx \text{Com}(\text{ck}, v')$.

A.2 Special Sound Protocols and Committed Transcripts

We refer to special-soundness and non-interactive arguments as defined by [AFK22].

Definition 12 (Public-coin interactive proof). An interactive proof $\Pi = (\text{P}, \text{V})$ for relation Ξ is an interactive protocol between two probabilistic machines, a prover P , and a polynomial time verifier V . Both P and V take as public input a statement $\mathbf{x}$ and, additionally, P takes as private input a witness $\mathbf{w}$ such that $(\mathbf{x}, \mathbf{w}) \in \Xi$. The verifier V outputs 1 if it accepts and 0 otherwise. Its output is denoted by $(\text{P}(\mathbf{x}, \mathbf{w}), \text{V}(\mathbf{x}))$. Accordingly, we say the corresponding transcript is accepting or rejecting. The protocol is public coin if the verifier randomness is public. The verifier messages are referred to as challenges. Π is a $(2\mu + 1)$ -move protocol if there are $\mu + 1$ prover messages a_i and μ verifier messages c_i . We refer to the transcripts as:

$$\text{tr} = (a_0, c_1, a_1, \dots, c_{\mu-1}, a_{\mu})$$

In this work, we will often refer to transcripts of special-sound protocols as *witnesses*, and simply denote them by $\mathbf{w}$. We do so following the line of work started by Protostar [BC23], which shows a family of special-sound protocols to prove different arithmetized relations that are used in common SNARKs. Our folding schemes will actually fold these transcripts.

Definition 13 (Special Soundness). Let Π be a $(2\mu + 1)$ -move public-coin interactive proof Π for relation Ξ , where the verifier samples its challenges from a set of size N . We say that Π is $(\alpha_1, \dots, \alpha_\mu)$ -out-of- N special sound if there exists a PPT extractor $\mathcal{E}$ such that, given any $(\alpha_1, \dots, \alpha_\mu)$ -tree of accepting transcripts for a statement $\mathbf{x}$, $\mathcal{E}$ outputs a witness $\mathbf{w}$ such that $(\mathbf{x}, \mathbf{w}) \in \Xi$ with overwhelming probability.

Random-Oracle Non-Interactive Argument of Knowledge. We say that a $(2\mu + 1)$ -move interactive protocol is *special-sound*, if it is α^μ -out-of- N special-sound, for $\alpha = \text{poly}(\lambda)$ and $N = \lambda^{\omega(1)}$. The results of [AFK22] imply that the non-interactive version of the protocol, after applying Fiat-Shamir transformation [FS87], is knowledge-sound with error $\text{negl}(\lambda)$ in the random oracle model.

For now on, we follow the conventions from Protostar [BC23, EG23] for committed transcripts and for committed relations which are consistent with respect to the Fiat-Shamir transformation. We introduce below the notions of committed transcripts and hash consistency used in [BC23, EG23].

Definition 14 (Committed Transcripts). For special-sound protocols with a transcript $\mathbf{w} = (a_0, c_1, a_1, \dots, c_{\mu-1}, a_\mu)$, we denote their committed transcript by

$$\mathbf{t} = \text{Com}(\mathbf{w}) = (\text{Com}(a_0), c_1, \text{Com}(a_1), \dots, c_{\mu-1}, \text{Com}(a_\mu))$$

where Com is used as shorthand for the operation of obtaining a committed transcript from a standard transcript.

We remark that if an interactive proof is special-sound, the committed version of the protocol (i.e. such that all prover messages are committed) is computationally special sound. The transformation holds by extending the definition of a valid extractor: for a committed special-sound protocol, given a valid tree of transcripts, an extractor should be able to extract *either a witness of the relation or a collision for the commitment scheme* with overwhelming probability, breaking commitment binding in the latter case. We refer to [EG23] for details and a formal exposition.

Finally, we can transform committed interactive special-sound proofs into non-interactive ones via the Fiat-Shamir transform. We say that a transcript is hash-consistent if it has been obtained by a correct application of the Fiat-Shamir transform, as follows:

Definition 15 (Hash consistency). Let $H(\cdot)$ be a hash function mapping to the challenge space and modeled as a random oracle. We say that a μ -round committed transcript $\mathbf{t}$ is $\mathcal{H}$ -consistent if:

- Let $\mathbf{t}_i = (\text{Com}(a_0), c_1, \text{Com}(a_1), \dots, \text{Com}(a_{i-1}))$ be the partial (committed) transcript of $\mathbf{t}$ up to the i -th round.
- For every $i \in [\mu]$, it holds that $c_i = H(\mathbf{t}_i)$.

For simplicity, we simply write $\mathcal{H}(\mathbf{t}) = 1$ if $\mathbf{t}$ is $\mathcal{H}$ consistent for the random oracle H .

B Full Description of Protogalaxy

B.1 Protogalaxy Description

Protogalaxy [EG23] folds a set of k witnesses $\mathbf{w}_1, \dots, \mathbf{w}_k$ that satisfy a base polynomial relation defined by

$$\Xi := \left\{ (\mathbf{t}, \mathbf{w}) \mid \mathbf{t} = \text{Com}(\mathbf{w}) \wedge f(\mathbf{w}) = \mathbf{0}_n \wedge \mathcal{H}(\mathbf{t}) = 1 \right\}. \quad (4)$$

for some polynomial map $f = (f_1, \dots, f_n) : \mathbb{F}^M \rightarrow \mathbb{F}^n$ of degree d defined over a finite field $\mathbb{F}$. We recall that whenever we write $\mathbf{t} = \text{Com}(\mathbf{w})$, we refer to the committed relations introduced in Definition 14 and the hash consistency check $\mathcal{H}(\mathbf{t}) = 1$ is as stated in Definition 15.

We present a description of Protogalaxy, which we generalize to work over arbitrary Lagrange domains $\mathbb{H} \subset \mathbb{F}$ that are not necessarily a subgroup.⁹ We fix positive integers $d, k < n$, and denote $t := \log n$. We also let $L_0(X), L_1(X), \dots, L_k(X)$ be the Lagrange basis over $\mathbb{H}$, and $Z(X) := \prod_{a \in \mathbb{H}} (X - a)$ be the corresponding vanishing polynomial. We present the protocol in Figure 4 and state it as a reduction of knowledge below. For the proof, we refer to [EG23].

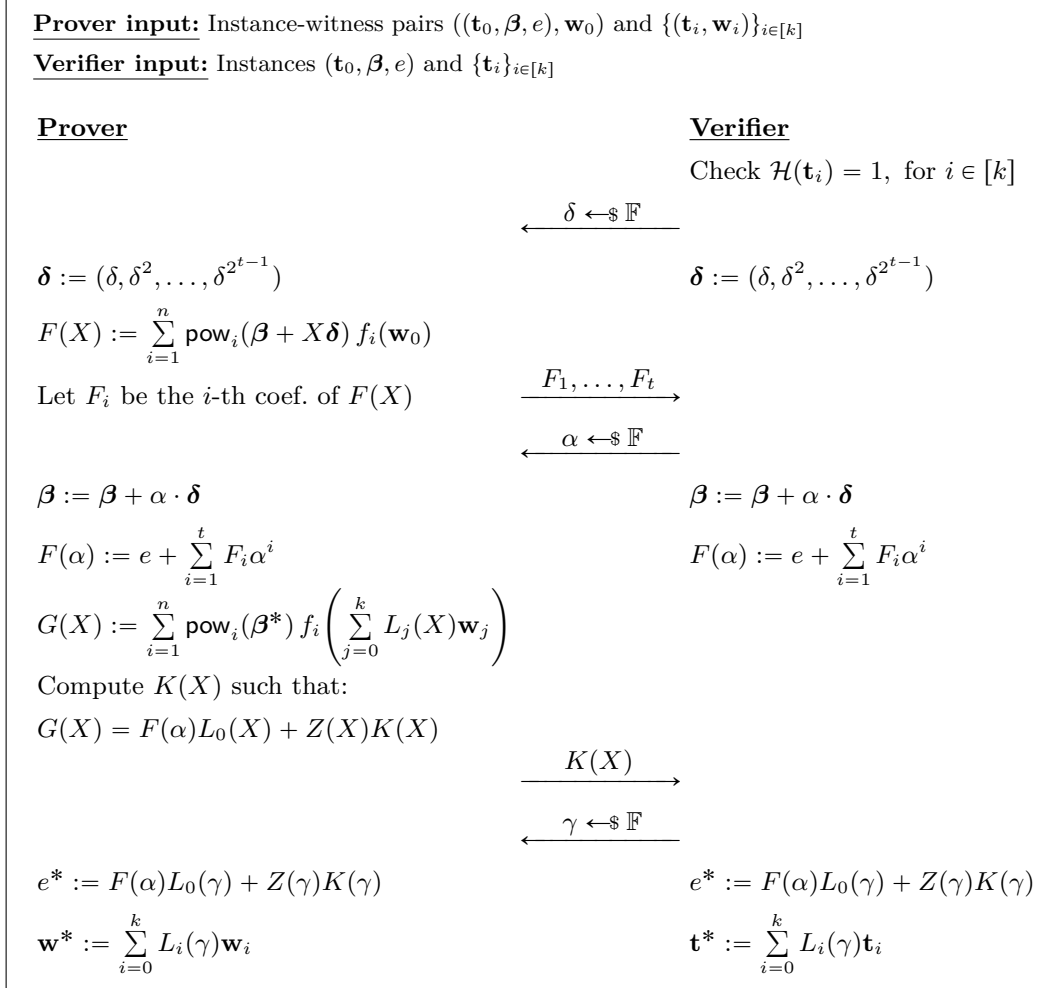


Fig. 4. Protogalaxy folding protocol $\Xi_{\text{rand}} \times \Xi^k \rightarrow \Xi_{\text{rand}}$

Theorem 4. *The protocol in Figure 4 is a reduction of knowledge for the relations $\Xi_{\text{rand}} \times \Xi^k \rightarrow \Xi_{\text{rand}}$.*

⁹ The most efficient instantiation, however, is when $\mathbb{H}$ is indeed the subgroup of roots of unity of $\mathbb{F}$ of order $k+1$ for a power-of-two $k+1$, due to the advantage of efficient FFT-based polynomial arithmetic.

C A Standalone Range Proof from Neo

In this section, we present the range proof from Neo/SuperNeo [NS26] as a standalone component, so that it can be plugged into a folding scheme such as ProtogaLattice independently.¹⁰ We also carry out a performance comparison, particularly on proof size, with the range proof of Latticefold [BC25a]. We remark that Cyclo [GLLO26] uses the same range proof as Neo.

Neo/SuperNeo embeds a norm check inside a larger batched sumcheck that simultaneously handles CCS constraints and accumulated evaluation claims. Strictly speaking, the protocol is a *reduction*: it converts a claim on the bound of the witnesses into an evaluation of multilinear extensions.

C.1 Notation and Relations

Let q be a prime and $\nu \geq 1$ an integer. The witness is a vector $\mathbf{w} \in \mathbb{F}_q^M$ over the base field, and all challenges are drawn from the extension $\mathbb{F}_{q^\nu} \supseteq \mathbb{F}_q$, chosen so that $1/q^\nu = \text{negl}(\lambda)$. In practice, $\nu = 2$ suffices for the Schwartz-Zippel bounds.

We assume that M is a power of two, padding witness vectors with zeroes whenever necessary, and let $\ell := \log_2 M$ denote the number of sumcheck rounds.

Polynomials. For $\ell \geq 1$ let $\text{eq}(x, y) := \prod_{i=1}^{\ell} (x_i y_i + (1 - x_i)(1 - y_i))$. For $v \in \mathbb{F}_q^M$ we write $\tilde{v}$ for the multilinear extension of v , evaluated at points of $\mathbb{F}_{q^\nu}^\ell$ through the inclusion $\mathbb{F}_q \subseteq \mathbb{F}_{q^\nu}$:

$$\tilde{v}(\mathbf{r}) = \sum_{\mathbf{x} \in \{0,1\}^\ell} \text{eq}(\mathbf{r}, \mathbf{x}) v_{\mathbf{x}}.$$

Relations. Fix b, k and M , and let Com be a binding commitment on $\mathbb{F}_q^M$. The range check is a reduction of knowledge $\Xi_{\text{norm}} \rightarrow \Xi_{\text{eval}}$, where the relations are defined as follows

$$\Xi_{\text{norm}} := \left\{ ((\mathbf{t}_i)_{i \in [k]}; (\mathbf{w}_i)_{i \in [k]}) : \mathbf{w}_i \in \mathbb{F}_q^M, \mathbf{t}_i = \text{Com}(\mathbf{w}_i), \|\mathbf{w}_i\|_\infty < b \right\}$$

to

$$\Xi_{\text{eval}} := \left\{ ((\mathbf{t}_i)_i, \mathbf{r}' \in \mathbb{F}_{q^\nu}^\ell, (y'_i)_i; (\mathbf{w}_i)_i) : \mathbf{t}_i = \text{Com}(\mathbf{w}_i), y'_i = \tilde{\mathbf{w}}_i(\mathbf{r}') \right\}.$$

Note that the commitments are not modified throughout the reduction of knowledge.

C.2 Protocol

The protocol is simply exploiting the following identity over $\mathbb{F}_q$,

$$\|x\|_\infty < b \iff \prod_{a=-b+1}^{b-1} (x - a) = 0. \quad (5)$$

¹⁰ We have used generative AI tools in this section to import notation and equations from [NS26], as well as to elaborate tables and protocol diagrams.

Applied coordinate-wise to $\mathbf{w} \in \mathbb{F}_q^M$ and lifted to multilinear extensions, the condition is a standard zero-check that we can express as a sumcheck. Moreover, the k instances are batched with powers of a single challenge $\gamma \in \mathbb{F}_{q^\nu}$. This yields:

$$\text{NC}(\mathbf{X}) := \sum_{i=1}^k \gamma^{i-1} \cdot \prod_{a=-b+1}^{b-1} \left(\widetilde{\mathbf{w}}_i(\mathbf{X}) - a \right), \quad \mathbf{X} := (X_1, \dots, X_\ell). \quad (6)$$

Note that the polynomial has coefficients in $\mathbb{F}_{q^\nu}$. This is because γ needs to come from the extension field for soundness.

Protocol Π_{NC}

Input: $((\mathbf{t}_i)_i; (\mathbf{w}_i)_{i \in [k]})$ with $\mathbf{w}_i \in \mathbb{F}_q^M$

Output: $((\mathbf{t}_i)_i, \mathbf{r}', (y'_i)_i; (\mathbf{w}_i)_i)$

1. $\mathcal{V}$: sample and send $\alpha \leftarrow \mathbb{F}_{q^\nu}^\ell$ and $\gamma \leftarrow \mathbb{F}_{q^\nu}$.
2. $\mathcal{P} \leftrightarrow \mathcal{V}$: define $Q(\mathbf{X}) := \text{eq}(\mathbf{X}, \alpha) \cdot \text{NC}(\mathbf{X}) \in \mathbb{F}_{q^\nu}[\mathbf{X}]$ for NC as in Equation (6). Run $\text{SumCheck}(0; Q)$ over $\{0, 1\}^\ell$, reducing $0 = \sum_{\mathbf{x} \in \{0, 1\}^\ell} Q(\mathbf{x})$ to an evaluation claim $v \stackrel{?}{=} Q(\mathbf{r}')$ at $\mathbf{r}' \leftarrow \mathbb{F}_{q^\nu}^\ell$.
3. $\mathcal{P}$: send $y'_i \leftarrow \widetilde{\mathbf{w}}_i(\mathbf{r}') \in \mathbb{F}_{q^\nu}$ for all $i \in [k]$.
4. $\mathcal{V}$: derive $\theta := \sum_{i=1}^k \gamma^{i-1} \prod_{a=-b+1}^{b-1} (y'_i - a)$ and accept if and only if $v \stackrel{?}{=} \text{eq}(\mathbf{r}', \alpha) \cdot \theta$.
5. Output $((\mathbf{t}_i)_i, \mathbf{r}', (y'_i)_i; (\mathbf{w}_i)_i)$.

The verifier's work in step 4 is $k(2b - 1)$ multiplications in $\mathbb{F}_{q^\nu}$ together with the $O(\ell)$ cost of evaluating $\text{eq}(\mathbf{r}', \alpha)$ and of the sumcheck verifier. For the completeness and soundness analysis, we defer the reader to [NS26].

C.3 Proof size for ternary vectors ($b = 2$)

Next, we analyze the proof size of the protocol. Setting $b = 2$ means $\|\mathbf{w}_i\|_\infty < 2$, that is, every coordinate lies in $\{-1, 0, 1\}$. This is the setting chosen by Neo, Latticefold and Cyclo. The sumcheck contributes ℓ rounds, in which a univariate polynomial of degree 4 is sent at each round. This can be done with four field elements sent per round (as one of them can be derived as a check). Step 3 contributes k elements. Since the challenges are not counted, we have that

$$|\pi_{\text{NC}}| = (4\ell + k) \cdot \nu \lceil \log_2 q \rceil \text{ bits}. \quad (7)$$

In Table 6, we give the resulting transcript sizes for parameters $\nu = 2$ and a 64-bit prime q . At the current parameter setting, the soundness analysis gives around 121 bits of security. To see this, observe that we are drawing challenges from a 128 bit-size extension field. The remaining 7 bits are lost due to the sumcheck itself.

Comparison with LatticeFold's range proof. LatticeFold [BC25a] checks norms with the polynomial

$$g_{\text{norm}}(\mathbf{x}) := \sum_{i=1}^{2k_{\text{dec}}} \mu_i \cdot \text{eq}(\beta, \mathbf{x}) \cdot \prod_{j=-(b-1)}^{b-1} \left(\widetilde{f}_i(\mathbf{x}) - j \right),$$

which is essentially identical to NC of Equation (6). The difference with Neo is that LatticeFold runs the sumcheck over the cyclotomic ring $\mathcal{R}_q$, with challenges from a strong sampling set $\mathcal{C} \subseteq \mathcal{R}_q$,

M	ℓ	$k = 1$	$k = 2$	$k = 3$	$k = 4$
2^{16}	16	1.02	1.03	1.05	1.06
2^{20}	20	1.27	1.28	1.30	1.31
2^{24}	24	1.52	1.53	1.55	1.56
2^{28}	28	1.77	1.78	1.80	1.81
2^{30}	30	1.89	1.91	1.92	1.94

Table 6. Transcript size of the range check Π_{NC} in KB, at $b = 2$, $\nu = 2$ and a 64-bit prime q .

so each round message is a tuple of $\mathcal{R}_q$ -elements. Neo runs the sumcheck over the field $\mathbb{F}_{q^\nu}$, resulting into much smaller communication.¹¹ To give a precise estimate, at $\ell = 26$ rounds, we have that LatticeFold’s range check is 52 KB, whereas Neo’s range check is 1.6 KB.

Adapting the range proof to ProtogaLattice. A ProtogaLattice witness is a ring vector, which we write $\mathbf{w}_{\mathcal{R}} \in \mathcal{R}_q^m$ to distinguish it from the field vector $\mathbf{w}$ of this appendix. Since $\|\mathbf{w}_{\mathcal{R}}\|_\infty$ is by definition the ℓ_∞ norm of its coefficient vector, Π_{NC} applies seamlessly with $\mathbf{w} := \text{cf}(\mathbf{w}_{\mathcal{R}}) \in \mathbb{F}_q^{Nm}$. Observe that its output fits our Definition 7 almost directly: the evaluation point and the claimed values become the public input $\mathbf{x}_{\text{RP}} := (\mathbf{r}', (y'_i)_i)$, while the k claims $y'_i = \widetilde{\mathbf{w}}_i(\mathbf{r}')$ are linear in the witness, so f_{RP} has degree one. The remaining step is to express the (extension) field evaluation $\widetilde{\mathbf{w}}(\mathbf{r}')$ as an $\mathcal{R}_q$ -constraint, to be able to fold it using ProtogaLattice. There are two ways to do this. One is to use the Neo embedding (see [NS26, Theorems. 9–10]), which is exactly the additional reduction of knowledge that we mentioned in Section 4, to translate field to ring constraints. If the range proof of Neo is used, that reduction of knowledge comes at minimal extra cost as the sumcheck can be batched with that of the range proof. The second way is to embed the extension field as a CRT slot of $\mathcal{R}_q$, which enables us to translate the sumcheck claim to the ring straightforwardly.

The range proof from LatticeFold can be also used straightforwardly by ProtogaLattice, but at a significantly higher overall cost.

D Deferred Proofs

D.1 Gröbner Bases over Rings

We consider the theory of Gröbner bases over a commutative noetherian ring $\mathcal{R}$ from [Pau07]. Let M be the polynomial ring $\mathcal{R}[X_1, \dots, X_k]$, viewed as a free $\mathcal{R}$ -module with the basis of monomials $(X^\alpha)_{\alpha \in \mathbb{N}^k}$. We implicitly consider a total order $\leq$ over the monomials of M . Throughout this document, for any non-zero polynomial $P = \sum_{\alpha \in \mathbb{N}^k} c_\alpha X^\alpha \in M$, we define its degree $\deg(P) = \max\{\alpha / c_\alpha \neq 0\}$, and $\text{lc}(P) = c_{\deg(P)}$ (respectively $\text{lm}(P) = \text{lc}(P)X^{\deg(P)}$) denotes the leading coefficient (respectively the leading monomial) of P . Finally, for a finite subset $F \subset M$ (respectively $F \subset \mathcal{R}$) we let $\langle F \rangle_M$ (respectively $\langle F \rangle_{\mathcal{R}}$) denote the ideal of M generated by the polynomials in F (respectively the ideal of $\mathcal{R}$ generated by F). We omit the subscript and write $\langle F \rangle$ when it is clear from context.

¹¹ Actually, a sumcheck over $\mathcal{R}_q$ can be seen as N/τ parallel sumchecks over $\mathbb{F}_{q^\tau}$ [BC25a, Sec. 5], so one ring transcript proves $N/\tau = 16$ field claims at once. This closes the gap between the two techniques significantly, but Neo’s proof is still more efficient.

Proposition 2 (Existence of division in M , [Pau07], Proposition 2). *Let F be a finite subset of $M \setminus \{0\}$ and $P \in M$. For all $P \in M$, there exists a remainder $R \in M$ and a tuple of quotients $(Q_f)_{f \in F} \in M^F$ such that:*

- $P = \sum_{f \in F} Q_f f + R$
- For all $f \in F$, $Q_f = 0$ or $\deg(Q_f) + \deg(f) \leq \deg(P)$
- $R = 0$ or $\text{lc}(R) \notin \langle \text{lc}(f)/f \in F \text{ such that } \deg(R) \in \deg(f) + \mathbb{N}^k \rangle_{\mathcal{R}}$.

For an ideal $\mathcal{I} \subset M$ and $\alpha \in \mathbb{N}^k$, we define the leading coefficient ideal $\text{lc}(\alpha, \mathcal{I}) = \langle \text{lc}(f)/f \in \mathcal{I} \text{ such that } \deg(f) = \alpha \rangle$.

Definition 16 (Gröbner basis, [Pau07], Definition 4). *A finite subset $\mathcal{G} \subset \mathcal{I} \setminus \{0\}$ is a Gröbner basis of $\mathcal{I}$ if and only if for all $\alpha \in \mathbb{N}^k$, the ideal $\text{lc}(\alpha, \mathcal{I})$ is generated by $\{\text{lc}(G)/G \in \mathcal{G}, \alpha \in \deg(G) + \mathbb{N}^k\}$.*

Proposition 3 ([Pau07], Proposition 7). *Let $\mathcal{G}$ be a Gröbner basis of $\mathcal{I}$. Then $P \in \mathcal{I}$ if and only if a remainder of P after division by $\mathcal{G}$ is 0.*

The following proposition provides a generalization of Buchberger's criterion, which also induces an algorithm for computing Gröbner bases over rings.

Proposition 4 (Gröbner basis criterion, [Pau07] Proposition 9). *Let $\mathcal{G}$ be a finite subset of $M \setminus \{0\}$, and let $\mathcal{I} := \langle G \rangle_M$. For any non-empty subset $E \subset \mathcal{G}$, let S_E be a finite set of generators of the $\mathcal{R}$ -module*

$$\left\{ (c_e)_{e \in E} \left| \sum_{e \in E} c_e \text{lc}(e) = 0 \right. \right\}$$

of syzygies of $(\text{lc}(e))$. Then, the following assertions are equivalent:

1. $\mathcal{G}$ is a Gröbner basis for $\mathcal{I}$
2. For all $E \subset \mathcal{G}$ and for all $(c_e)_{e \in E} \in S_E$ a remainder of

$$\sum_{e \in E} c_e X^{m - \deg(e)} e,$$

where $m = (\max_{e \in E} (\deg(e)_i))_{i \in [k]}$ after division by $\mathcal{G}$ is 0.

Reduced Gröbner bases. To ensure the uniqueness of the remainder, we rely on the notion of reduced Gröbner bases. We assume the existence of a map Gen that takes an ideal $\mathcal{J} \subset \mathcal{R}$ and returns a system of canonical generators $\text{Gen}(\mathcal{J})$ of $\mathcal{J}$. Additionally, we fix a set of canonical representatives for the coefficients: For any ideal $\mathcal{J} \subset \mathcal{R}$ and any coset $z + \mathcal{J} \subset \mathcal{R}$, the function R selects a unique element $R(z, \mathcal{J})$. We require $R(0, \mathcal{J}) = 0$.

Definition 17 (Reduced Gröbner bases, [Pau07], Definition 17). *Let $\mathcal{I}$ be an ideal of M . A Gröbner basis $\mathcal{G}$ of $\mathcal{I}$ is a reduced Gröbner basis if and only iff:*

- For all $\alpha \in \deg(\mathcal{I})$ the map $G \mapsto \text{lc}(G)$ from $\{G \in \mathcal{G} / \deg(G) = \alpha\}$ to $\{R(z, \text{lc}(\alpha, \mathcal{I})) / z \in \text{Gen}(\text{lc}(\alpha, \mathcal{I}) \setminus \{0\})\}$ is bijective
- For all $G := \sum_{\beta \in \mathbb{N}^k} c_\beta X^\beta \in \mathcal{G}$ and for all $\alpha \in \mathbb{N}^k$ with $c_\alpha \neq 0$ and $\alpha \neq \deg(G)$, we have $c_\alpha = R(c_\alpha, \text{lc}(\alpha, \mathcal{I}))$.

Proposition 5 (Uniqueness, extension of [Pau07] Proposition 19). *Every ideal $\mathcal{I}$ in M has a unique reduced Gröbner basis. In particular, up to the choice of the maps Gen and $\mathcal{R}$, the remainder in the division by the reduced Gröbner basis of $\mathcal{I}$ is unique.*

D.2 Proof of Theorem 1

Proof. We split the proof in 4 claims.

CLAIM 1: The subset of invertible-or-zero points in the variety of $\mathcal{I}$ (i.e the set of common zeros of all polynomials of $\mathcal{I}$) is the set of $k + 1$ points $\{P_j, j \in [0, k]\}$, where P_j 's last j coefficients are ones and the others are zeroes.

Let us find the zeros of $\mathcal{I}$ with invertible-or-zero coefficients, first considering all k equations $Z_{jj} = Y_j(1 - Y_j) = 0$, with $j \in [0, k - 1]$. Since we assume that all Y_j 's and $1 - Y_j$'s are invertible or zero, we have that $\forall j \in [0, k - 1], Y_j \in \{0, 1\}$.

Initialization: If $k = 1$, the invertible-or-zero points in the variety are the two points $Y_0 = 0$ and $Y_0 = 1$.

Induction: Assume that the variety of $\mathcal{I}$ is indeed the set of $k + 1$ points $\{P_j, j \in [0, k]\}$, where P_j 's last j coefficients are ones and the others are zeros. We consider the next case for $k + 1$ variables. Let $\mathcal{I}' := \langle (Z_{j+1,j})_{j \in [0, k]} \rangle + \langle (Z_{jj})_{j \in [0, k]} \rangle$. Again, we have $Y_j \in \{0, 1\}$ for all $j \in [0, k]$. If $Y_k = 0$, it follows from $Y_{k-1}(1 - Y_k) = 0$ that $Y_{k-1} = 0$, thus by the induction hypothesis, the only one point for which $Y_k = 0$ is $(0, \dots, 0)$. If $Y_k = 1$, then we have no additional constraint on Y_{k-1} , thus the concatenation of any point P_j of $\mathcal{I}$ and 1 is a point of $\mathcal{I}'$. In the end, the invertible or zero points in the variety of $\mathcal{I}'$ is the set of $k + 2$ points of the next induction step:

$$\mathcal{I}' = \{(0, \dots, 0) \text{ and } \forall j \in [0, k], (P_j || 1)\}.$$

CLAIM 2: The polynomial $w_k^*(\mathcal{Y})$ interpolates the w_j 's over the invertible or zero points of the variety of $\mathcal{I}$, i.e $w_k^*(P_j) = w_j$.

Recall that via Lemma 4,

$$w_k^*(\mathcal{Y}) = Y_0 w_0 + \sum_{j=1}^k (Y_j - Y_{j-1}) w_j.$$

Define polynomials c_j such that $w_k^* = \sum_{j=1}^k c_j w_j$, i.e $c_0 = Y_0$, $c_k = 1 - Y_{k-1}$ and for all $j \in [0, k - 1] c_j = Y_j - Y_{j-1}$. Since Y_0 is zero across all the P_j 's, except for the last $P_k = (1, \dots, 1)$, we have $c_0(P_j) = 0$ for all $j \in [0, k - 1]$ and $c_0(P_k) = 1$. Conversely, $c_k(P_j) = 1$ for all $j \in [k]$ and $c_k(P_0) = 0$. For all other coefficients c_j for $j \in [k - 1]$, we have that $c_j(Y_0, \dots, Y_{k-1}) = Y_j - Y_{j-1}$. Consecutive coordinates of P_j are equal, except when $\ell = j$, thus $c_j(P_\ell) = 0$ for all ℓ except $\ell = j$ where $c_j(P_j) = 1$. It follows that $\forall i, j \in [0, k] c_i(P_j) = 0$ if $i \neq j$ and $c_j(P_j) = 1$.

CLAIM 3: We consider the quotient $\mathcal{M} = \mathcal{R}_q[Y_0, \dots, Y_{k-1}]$, and prove that the map $\phi : \mathcal{M} \rightarrow \mathcal{R}_q^{k+1}$ defined by

$$\phi(P) \mapsto (P(P_j))_{j \in [0, k]}$$

is an isomorphism of $\mathcal{R}_q$ -modules.

We consider the basis $B = (Y_0, \dots, Y_{k-1}, 1)$ of $\mathcal{M}/(\mathcal{I})$. To justify that B is indeed a basis of $\mathcal{M}/(\mathcal{I})$, consider any monomial $U = Y_0^{a_0} \dots Y_{k-1}^{a_{k-1}}$, and notice that via the polynomials $Z_{00}, \dots, Z_{k-1, k-1} \in \mathcal{I}$, we have $U = \prod_{i \text{ st } a_i \neq 0} Y_i$. Next, because $Z_{j+1, j} = 0$, we have $Y_{j+1} = Y_j Y_{j+1}$, thus by induction we obtain $U = Y_{i_{\max}}$ with $i_{\max} = \max\{i, a_i \neq 0\}$ or $U = 1$ if $\{i, a_i \neq 0\}$ is empty. It follows that B is, indeed, a basis for $\mathcal{M}/\mathcal{I}$, thus the dimension of $\mathcal{M}/\mathcal{I}$ is $k + 1$.

We now calculate the matrix of ϕ in basis B to basis B . We have $b_i(P_j) = 0$ for all $i < j$ and 1 otherwise. It follows that the matrix of ϕ is anti-triangular, with ones on the diagonal, thus invertible, hence ϕ is an isomorphism.

CLAIM 4: Putting the pieces together, we show that

$$f\left(Y_0w_0 + \sum_{j=1}^k (Y_j - Y_{j-1})w_j\right) - \left(Y_0f(w_0) + \sum_{j=1}^k (Y_j - Y_{j-1})f(w_j)\right) \in \mathcal{I}.$$

We recall that from claim 2, we have that the polynomial

$$\tilde{H}(\mathcal{Y}) := f\left(Y_0w_0 + \sum_{j=1}^k (Y_j - Y_{j-1})w_j\right) - \left(Y_0f(w_0) + \sum_{j=1}^k (Y_j - Y_{j-1})f(w_j)\right)$$

vanishes on the P_j 's, thus $\phi(\tilde{H}) = 0$. Since from claim 3, ϕ is an isomorphism, we have $\tilde{H} \bmod \mathcal{I} = 0$ i.e $\tilde{H} \in \mathcal{I}$ and the result follows. $\square$

D.3 Proof of Proposition 1

Proof. To prove that $\mathcal{G}$ is a Gröbner basis for $\mathcal{I}$, we use the extension of Buchberger's criterion to rings Proposition 4. Since all leading coefficients $\text{lc}(\mathcal{G})$ are 1 (and thus units of $\mathcal{R}_q$), for any subset $E \subset \mathcal{G}$, the syzygy module of the leading coefficients of E is:

$$\sum_{e \in E} c_e = 0.$$

This module is generated by vectors with one 1, one -1 and zeros elsewhere. Hence the generalized criterion from Proposition 4 reduces to the usual Buchberger S -polynomial criterion, i.e for all $G_i, G_j \in \mathcal{G}$, with $g_i = \text{lm}(g_i)$ and $g_j = \text{lm}(g_j)$, a remainder of

$$S(G_i, G_j) = \frac{\text{lcm}(g_i, g_j)}{g_i} G_i - \frac{\text{lcm}(g_i, g_j)}{g_j} G_j$$

is 0.

We now proceed to show that for all S -polynomials of pairs of elements of $\mathcal{G}$ have remainder 0 in the division by $\mathcal{G}$. Consider two polynomials Z_{ij}, Z_{rs} , with $i, j, r, t \in [0, k], i \geq j$ and $r \geq t$.

The first case is when $\{i, j\} \cap \{r, s\} = \emptyset$. In this case, Z_{ij} and Z_{rs} are coprime, thus $S(Z_{ij}, Z_{rs}) \in \langle \mathcal{G} \rangle$.

When $i = r$, we have

$$\begin{aligned} S(Z_{ij}, Z_{is}) &= \frac{Y_i Y_j Y_s}{Y_i Y_j} Z_{ij} - \frac{Y_i Y_j Y_s}{Y_i Y_s} Z_{is} \\ &= Y_s(Y_i Y_j - Y_i) - Y_j(Y_i Y_s - Y_i) \\ &= Y_i Y_j - Y_i Y_s \\ &= (Y_i Y_j - Y_i) - (Y_i Y_s - Y_i) \\ &= Z_{ij} - Z_{is}, \end{aligned}$$

thus the remainder is 0.

When $i = s$, we have

$$\begin{aligned}
S(Z_{ij}, Z_{ri}) &= \frac{Y_i Y_j Y_r}{Y_i Y_j} Z_{ij} - \frac{Y_i Y_j Y_r}{Y_i Y_r} Z_{ri} \\
&= Y_r(Y_i Y_j - Y_i) - Y_j(Y_r Y_i - Y_r) \\
&= Y_r Y_i - Y_r Y_j \\
&= (Y_r Y_i - Y_r) - (Y_r Y_j - Y_r) \\
&= Z_{ri} - Z_{rj},
\end{aligned}$$

therefore the remainder is 0. Notice that since $i \geq j$ and $r \geq i$, we have $r \geq j$ thus $Z_{rj} \in \mathcal{G}$.

When $j = s$, we have

$$\begin{aligned}
S(Z_{ij}, Z_{rj}) &= \frac{Y_i Y_j Y_r}{Y_i Y_j} Z_{ij} - \frac{Y_i Y_j Y_r}{Y_r Y_j} Z_{rj} \\
&= Y_r(Y_i Y_j - Y_i) - Y_i(Y_r Y_j - Y_r) \\
&= 0.
\end{aligned}$$

We now prove that $\mathcal{G}$ is a reduced Gröbner basis. Since all the leading coefficients of the Z_{ij} 's are 1, we trivially chose $\mathbf{Gen}$ to map to $\{1\}$. Furthermore, since all the leading coefficients are 1, we only need to define $R(z, \langle 0 \rangle_{\mathcal{R}_q})$, thus $R = 0$. With both $\mathbf{Gen}$ and R being trivial, the conditions for $\mathcal{G}$ to be a reduced Gröbner basis for $\mathcal{I}$ w.r.t Definition 17 become the usual conditions for a Gröbner basis to be reduced over fields, that is,

1. For all $G \in \mathcal{G}$, $\text{lc}(G) = 1$
2. For all $G \in \mathcal{G}$, no monomial of G is in $\langle \text{lm}(G') | G' \in \mathcal{G} \setminus \{G\} \rangle$

We do have that the leading coefficient of Z_{ij} is 1 for all $i \geq j$, so we now prove 2). Consider $Z_{ij} = Y_i Y_j - Y_i \in \mathcal{G}$. The leading monomials of $\mathcal{G}$ are of degree 2, therefore $Y_i \notin \langle \text{lm}(G') | G' \in \mathcal{G} \setminus \{G\} \rangle$. Moreover, the only monomial of degree 2 that divides $Y_i Y_j$ is itself. Since Z_{ji} is not in $\mathcal{G}$, we have that $Y_i Y_j$ is also not in $\langle \text{lm}(G') | G' \in \mathcal{G} \setminus \{G\} \rangle$, therefore 2) is satisfied, which completes the proof. $\square$

D.4 Correctness and Soundness of PGL-Fold: Proof of Theorem 2

*Proof. **Correctness:*** Consider an honestly generated transcript of Figure 2. More precisely, let $((\mathbf{t}_0, \mathbf{x}_0^{\text{RP}}, \beta, e), \mathbf{w}_0) \in \Xi_{f,n,\gamma,1}^{\text{acc}}$ be an input accumulator instance-witness pair and let $((\mathbf{t}_j, \mathbf{x}_j^{\text{RP}}), \mathbf{w}_j)_{j \in [k]} \in (\Xi_{f,n,B,1}^{\text{fold}})^k$ be k input instance-witness pairs.

We prove that the output $((\mathbf{t}^*, \mathbf{x}^{\text{RP},*}, \beta^*, e^*), \mathbf{w}^*)$ is in the relation $\Xi_{f,n,\gamma',1}^{\text{acc}}$, with $\gamma' = \gamma + (2Tk + 1)B$.

By definition of $\mathbf{w}^*, \mathbf{t}^*$ and the linearity of Ajtai commitments, we have that $\mathbf{t}^* = \sum_j c_j \mathbf{t}_j = \sum_j c_j \mathbf{A} \mathbf{w}_j = \mathbf{A}(\sum_j c_j \mathbf{w}_j)$. We check the norm bound on $\mathbf{w}^*$ to ensure that $\mathbf{t}^*$ is indeed an Ajtai commitment to $\mathbf{w}^*$:

$$\begin{aligned}
\|\mathbf{w}^*\| &= \left\| \sum_{j=0}^k L_j(\mathbf{y}) \mathbf{w}_j \right\| \\
&= \left\| \mathbf{w}_0 + (y_1 - 1) \mathbf{w}_1 + \sum_{j=2}^k (y_j - y_{j-1}) \mathbf{w}_j \right\| \\
&\leq \gamma + (T + 1)B + 2T(k - 1)B \\
&\leq \gamma + (2Tk + 1)B = \gamma'.
\end{aligned}$$

To finish the proof, we need to prove that $\sum_{i=1}^n \text{pow}_i(\beta^*) f_i(\mathbf{w}^*) = e^*$.

Let $i \in [n]$. Via Lemma 4, we can substitute $\mathbf{w}^*$ with its closed form:

$$\mathbf{w}^* = \left(\prod_{j=1}^k X_j \right) \mathbf{w}_0 + \sum_{j=1}^k \left((1 - X_j) \prod_{\ell=j+1}^k X_\ell \right) \mathbf{w}_j$$

in the expression of $f_i(\mathbf{w}^*)$. We have:

$$f_i(\mathbf{w}^*) = f_i \left[\left(\prod_{j=1}^k X_j \right) \mathbf{w}_0 + \sum_{j=1}^k \left((1 - X_j) \prod_{\ell=j+1}^k X_\ell \right) \mathbf{w}_j \right] \quad (8)$$

$$= \left(Y_0 f_i(\mathbf{w}_0) + \sum_{j=1}^k (Y_j - Y_{j-1}) f_i(\mathbf{w}_j) \right) \mod \mathcal{I} \quad (9)$$

$$= Y_0 f_i(\mathbf{w}_0) + \sum_{j=1}^k (Y_j - Y_{j-1}) f_i(\mathbf{w}_j) + \sum_{1 \leq r \leq s \leq k} Q_{rs}^{(i)}(\mathcal{Y}) Z_{rs}(\mathcal{Y}), \quad (10)$$

$$= f_i(\mathbf{w}_0) + \sum_{1 \leq r \leq s \leq k} Q_{rs}^{(i)}(\mathbf{y}) Z_{rs}(\mathbf{y}) \quad (11)$$

where Equation (8) follows from Lemma 4, Equation (9) from Theorem 1, and Equation (10) follows from the existence of the quotient in Proposition 1 (they are unique up to syzygies). Finally, Equation (11) follows from the fact that each $\mathbf{w}_j$ with $j \in [k]$ is a witness of $\Xi_{f,n,B,1}^{\text{fold}}$, so $f_i(\mathbf{w}_j) = 0$ for every $i \in [n]$.

Since the Q_{rs} and the K_{rs} are equal up to syzygies, any other choice of a quotient gives the same value of $\sum_{r \leq s} K_{rs} Z_{rs}$. Multiplying Equation (11) by $\text{pow}_i(\beta^*)$ and summing over i ,

$$\begin{aligned}
\sum_{i=1}^n \text{pow}_i(\beta^*) f_i(\mathbf{w}^*) &= \sum_{i=1}^n \text{pow}_i(\beta^*) f_i(\mathbf{w}_0) + \sum_{1 \leq r \leq s \leq k} K_{rs}(\mathbf{y}) Z_{rs}(\mathbf{y}) \\
&= F(\alpha) + \sum_{1 \leq r \leq s \leq k} K_{rs}(\mathbf{y}) Z_{rs}(\mathbf{y}) = e^*.
\end{aligned}$$

Soundness.

We define an extractor $\mathcal{E}$ that given rewindable black-box access to a prover $\mathcal{A}$ returns either a collision for the commitment scheme or tuples $(\mathbf{t}_j, \mathbf{w}_j)_{j \in [k]} \in (\Xi_{f,n,B_{\text{ext}},\varrho}^{\text{fold}})^k$ and $((\mathbf{t}_0, \beta, e), \mathbf{w}_0) \in \Xi_{f,n,B_{\text{ext}},\varrho}^{\text{acc}}$.

Extraction : The transcripts provided by the prover are of the form

$$((\mathbf{t}_j)_{j \in [0,k]}, \beta, e, \delta, (F_i)_{i \in [t]}, \alpha, (K_{rs})_{r,s}, \mathbf{y}, \mathbf{t}^*, \beta^*, e^*, \mathbf{w}^*).$$

Our extractor $\mathcal{E}$ does the following:

1. Run $\mathcal{A}$ and obtain a first transcript

$$((\mathbf{t}_j)_{j \in [0,k]}, \beta, e, \delta, (F_i)_{i \in [t]}, \alpha, (K_{rs})_{r,s}, \mathbf{y}, \mathbf{t}^*, \beta^*, e^*, \mathbf{w}^*).$$

2. For all $j \in [k]$: Rewind the prover and send a new challenge $\mathbf{y}'_j = (1, y_1, \dots, y'_j, \dots, y_k)$, where y'_j is sampled independently from the same distribution $\mathcal{C}$ as y_j . Let $\mathbf{w}_j^*$ be the j -th accumulator and $\mathbf{t}_j^*$ be the accumulator's commitment.
3. For all $j \in [k]$, construct relaxed openings (Δ_j, z_j) to the commitment $(\mathbf{t}_j - \mathbf{t}_{j+1})$, with $\mathbf{t}_{k+1} := \mathbf{0}$, where $z_j := y_j - y'_j$ and $\Delta_j := z_j^{-1}(\mathbf{w}^* - \mathbf{w}_j^*)$.
4. Define $\bar{\mathbf{w}}_k := \Delta_k$ and $s_k := z_k$.
5. Define recursively for all $j = k - 1$ down to 1: $\bar{\mathbf{w}}_j := \bar{\mathbf{w}}_{j+1} + \Delta_j$ and $s_j := \prod_{i=j}^k z_i$.
6. Compute the accumulator opening: $\bar{\mathbf{w}}_0 := \mathbf{w}^* - \sum_{j=1}^k L_j(\mathbf{y}) \bar{\mathbf{w}}_j$ with slack $s_0 := s_1$.
7. The extractor outputs either the collision, or the witnesses $(\bar{\mathbf{w}}_j, s_j)_{j \in [0,k]}$.

We prove step by step that the extracted $(\bar{\mathbf{w}}_j)_{j \in [0,k]}$ are in the relation $(\Xi_{f,n,B_{\text{ext}},\varrho}^{\text{fold}})^k \times \Xi_{f,n,B_{\text{ext}},\varrho}^{\text{acc}}$, where $B_{\text{ext}} := (2T)^k(k^2 + k + 1)\gamma'$ and $\varrho = (2T)^k$.

For $j \in [k]$, (Δ_j, z_j) are relaxed openings for the commitments $\mathbf{t}_j - \mathbf{t}_{j+1}$.

We show that $(\Delta_j, z_j)_{j \in [k]}$ defined in step 3) are relaxed openings to the commitments $(\mathbf{t}_j - \mathbf{t}_{j+1})$. More precisely, they satisfy (i) $\mathbf{A}\Delta_j = \mathbf{t}_j - \mathbf{t}_{j+1}$, (ii) $\|z_j \Delta_j\| \leq 2\gamma'$ and (iii) $\|z_j\|_{\text{op}} \leq 2T$.

The output accumulators $(\mathbf{w}_j^*)_{j \in [k]}$ of all transcripts satisfy $\mathbf{A}\mathbf{w}_j^* = \mathbf{t}_j^*$. By construction of $\mathbf{t}^*$, we have

$$\begin{aligned} \mathbf{t}^* &= \sum_{j=0}^k L_j(\mathbf{y}) \mathbf{t}_j \\ &= \mathbf{t}_0 + \sum_{j=1}^k (y_j - y_{j-1}) \mathbf{t}_j \\ &= \mathbf{t}_0 + \sum_{j=1}^k y_j \mathbf{t}_j - \sum_{j=0}^{k-1} y_j \mathbf{t}_{j+1} \\ &= \mathbf{t}_0 - \mathbf{t}_1 + y_k \mathbf{t}_k + \sum_{j=1}^{k-1} y_j (\mathbf{t}_j - \mathbf{t}_{j+1}), \end{aligned}$$

therefore for all $j \in [k]$ we have $\mathbf{t}^* - \mathbf{t}_j^* = (y_j - y'_j)(\mathbf{t}_j - \mathbf{t}_{j+1}) = z_j(\mathbf{t}_j - \mathbf{t}_{j+1})$. It follows that $\mathbf{A}\Delta_j = \mathbf{A}(\mathbf{w}^* - \mathbf{w}_j^*) = (\mathbf{t}_j - \mathbf{t}_{j+1})$. To complete this proof, we bound the norm of $z_j \Delta_j$, which is exactly $\mathbf{w}^* - \mathbf{w}_j^*$, by the output accumulator bound γ' as follows

$$\|z_j \Delta_j\| = \|\mathbf{w}^* - \mathbf{w}_j^*\| \leq 2\gamma'.$$

For $j \in [k]$, $(\bar{\mathbf{w}}_j, s_j)$ are relaxed openings. We show that the extracted $(\bar{\mathbf{w}}_j, s_j)_{j \in [0, k]}$ are relaxed openings w.r.t the commitments $(\mathbf{t}_j)_{j \in [k]}$, i.e (i) $\mathbf{A}\bar{\mathbf{w}}_j = \mathbf{t}_j$ (ii) $\|s_j \bar{\mathbf{w}}_j\| \leq B_{\text{ext}}$ (iii) $\|s_j\|_{\text{op}} \leq \varrho$.

For $j = k$, the results hold because $\|s_k\|_{\text{op}} = \|y_k - y'_k\|_{\text{op}} \leq 2T \leq \varrho$,

$$\begin{aligned}\mathbf{A}\bar{\mathbf{w}}_k &= \mathbf{A}\Delta_k \\ &= \mathbf{t}_k,\end{aligned}$$

and

$$\begin{aligned}\|s_k \bar{\mathbf{w}}_k\| &= \|s_k \Delta_k\| \\ &\leq 2\gamma' \\ &< B_{\text{ext}}.\end{aligned}$$

For $j \in [1, k-1]$, work by induction from k down to j . For $\|s_j\|$, we have

$$\begin{aligned}\|s_j\|_{\text{op}} &= \|z_j \cdots z_k\|_{\text{op}} \\ &\leq \|z_j\|_{\text{op}} \cdots \|z_k\|_{\text{op}} \\ &\leq (2T)^{k-j+1} \\ &\leq \varrho.\end{aligned}$$

For commitments, we have

$$\begin{aligned}\mathbf{A}\bar{\mathbf{w}}_j &= \mathbf{A}(\bar{\mathbf{w}}_{j+1} + \Delta_j) \\ &= \mathbf{t}_{j+1} + \mathbf{t}_j - \mathbf{t}_{j+1} \\ &= \mathbf{t}_j.\end{aligned}$$

For the slack factor, we have as above

$$\begin{aligned}\|s_j\|_{\text{op}} &= \left\| \prod_{i=j}^k z_i \right\|_{\text{op}} \\ &\leq (2T)^{k-j+1} \\ &\leq \varrho.\end{aligned}$$

For the norm bound, we have

$$\begin{aligned}\|s_j \bar{\mathbf{w}}_j\| &= \|z_j s_{j+1} \bar{\mathbf{w}}_{j+1} + z_j s_{j+1} \Delta_j\| \\ &\leq 2T \|s_{j+1} \bar{\mathbf{w}}_{j+1}\| + (2T)^{k-j} \|z_j \Delta_j\|,\end{aligned}$$

using $s_j = z_j s_{j+1}$ and $\|s_{j+1}\|_{\text{op}} \leq (2T)^{k-j}$. Thus,

$$\begin{aligned}\|s_j \bar{\mathbf{w}}_j\| &\leq (k-j+1)(2T)^{k-j}(2\gamma') \\ &< B_{\text{ext}}.\end{aligned}$$

The extracted accumulator $\bar{\mathbf{w}}_0$ is a relaxed opening to $\mathbf{t}_0$. We now prove that $\bar{\mathbf{w}}_0$ satisfies (i), (ii) and (iii). For the slack factor, we have $s_0 = s_1$, thus $\|s_0\|_{\text{op}} \leq \varrho$. For the commitment, we have

$$\begin{aligned} \mathbf{A}\bar{\mathbf{w}}_0 &= \mathbf{A}(\mathbf{w}^* - \sum_{j=1}^k L_j(\mathbf{y})\bar{\mathbf{w}}_j) \\ &= \mathbf{t}^* - \sum_{j=1}^k L_j(\mathbf{y})\bar{\mathbf{t}}_j \\ &= \mathbf{t}_0. \end{aligned}$$

For the norm bound, we have

$$\begin{aligned} \|s_0\bar{\mathbf{w}}_0\| &= \|s_0\mathbf{w}^* - s_0 \sum_{j=1}^k L_j(\mathbf{y})\bar{\mathbf{w}}_j\| \\ &= \|s_0\mathbf{w}^* - \sum_{j=1}^k \frac{s_0}{s_j} \cdot L_j(\mathbf{y}) \cdot s_j\bar{\mathbf{w}}_j\| \\ &\leq (2T)^k \gamma' + \sum_{j=1}^k (2T)^{j-1} \cdot (2T) \cdot \|s_j\bar{\mathbf{w}}_j\| \\ &\leq (2T)^k \gamma' + \sum_{j=1}^k (2T)^j (2T)^{k-j} 2(k-j+1)\gamma' \\ &\leq (2T)^k \gamma' (1 + k(k+1)) \\ &\leq (2T)^k \gamma' (k^2 + k + 1) = B_{\text{ext}}. \end{aligned}$$

Under the hardness of $\text{MSIS}_{\kappa, m, 2\varrho B_{\text{ext}}}^{\infty, q}$, the $(\bar{\mathbf{w}}_j)_{j \in [0, k]}$ are unique.

The extracted witnesses satisfy the constraints. In the final part of the proof, we show that with

$$\Phi^*(X) = \sum_{i=1}^n \text{pow}_i(\beta^*) f_i(X) \quad \text{and} \quad \Phi(X) = \sum_{i=1}^n \text{pow}_i(\beta) f_i(X),$$

the following holds

$$(\bar{\omega}_j)_{j \in [k]} \quad \text{satisfies} \quad \Phi^*(\bar{\omega}_j) = 0 \quad \text{and} \quad \Phi^*(\bar{\omega}_0) = F(\alpha).$$

On the one hand, for any challenge vector submitted by the extractor, substituting $\mathbf{w}^*$ with $\sum_{j \in [0, k]} L_j(\mathbf{y})\bar{\omega}_j$ yields:

$$\begin{aligned} \Phi^*(\omega^*(\mathbf{y})) &= \Phi^*\left(\sum_{j \in [0, k]} L_j(\mathbf{y})\bar{\omega}_j\right) \\ &= \sum_{j=0}^k L_j(\mathbf{y})\Phi^*(\bar{\omega}_j) + \sum_{r,s} Q_{rs}(\mathbf{y})Z_{rs}(\mathbf{y}) \\ &= \Phi^*(\bar{\omega}_0) + \sum_{j=1}^k L_j(\mathbf{y})\Phi^*(\bar{\omega}_j) + \sum_{r,s} Q_{rs}(\mathbf{y})Z_{rs}(\mathbf{y}). \end{aligned}$$

On the other hand, via the verification equation, we have:

$$\begin{aligned}\Phi^*(\mathbf{w}^*(\mathbf{y})) &= e^* \\ &= \sum_{r,s} K_{rs}(\mathbf{y}) Z_{rs}(\mathbf{y}) + F(\alpha).\end{aligned}$$

Plugging both together, we have that the following polynomial:

$$\Psi(\mathcal{Y}) := (\Phi^*(\bar{\omega}_0) - F(\alpha)) + \sum_{j=1}^k L_j(\mathcal{Y}) \Phi^*(\bar{\omega}_j) + \sum_{r,s} (Q_{rs}(\mathcal{Y}) - K_{rs}(\mathcal{Y})) Z_{rs}(\mathcal{Y})$$

vanishes at all accepting transcripts. Under the Schwartz-Zippel Lemma over rings Lemma 1, the probability that Ψ is non-zero and vanishes for $\mathcal{Y} \leftarrow \mathcal{C}^k$ is bounded by

$$\mathbb{P}_{\mathcal{Y} \leftarrow \mathcal{C}^k}(\Psi(\mathcal{Y}) = 0) \leq \frac{\deg(\Psi)}{|\mathcal{C}|} \leq \frac{dk}{|\mathcal{C}|}.$$

It follows that with probability at least $1 - \frac{dk}{|\mathcal{C}|}$, we have $\Psi = 0$ as a polynomial. Under the uniqueness of the decomposition in the reduced Gröbner basis Propositions 1 and 5, it follows that

$$\forall j \in [k], \quad \Phi^*(\bar{\omega}_j) = 0 \tag{12}$$

$$\Phi^*(\bar{\omega}_0) = F(\alpha) \tag{13}$$

$$\sum_{r,s} K_{rs} Z_{rs} = \sum_{r,s} Q_{rs} Z_{rs}.$$

Both Equations (12) and (13) are polynomials in α, δ of total degree $2 \log n$ that vanish on the random points (α, β) drawn from $\mathcal{C}^2$, where we recall $\mathcal{C}$ is a strong sampling set. It follows that with probability at least $\frac{2 \log n}{|\mathcal{C}|}$, these polynomials are 0, thus by identifying each coefficient, we have:

$$\forall j \in [k], f(\bar{\omega}_j) = 0 \text{ and}$$

$$\sum_{i=1}^n \text{pow}_i(\beta) f_i(\bar{\omega}_0) = e,$$

which completes the proof.

D.5 Correctness and Soundness of PGL-Boot: Proof of Theorem 3

Proof. Correctness. Let $((\mathbf{t}, \mathbf{x}^{\text{RP}}, \beta, e), \mathbf{w}) \in \Xi_{f,n,\gamma',1}^{\text{acc}}$. Since $b^{k'} \geq \gamma'$, the $\mathbf{w}_j$'s are the base b decomposition of $\mathbf{w}$, i.e $\sum_{j=0}^{k'-1} b^j \mathbf{w}_j = \mathbf{w}$ and $\|\mathbf{w}_i\|_\infty \leq b - 1$. By the linearity of the commitment scheme, the first check $\sum_{j=0}^{k'-1} b^j \mathbf{t}_j$ passes. The second check is a direct consequence of Theorem 1 evaluated at $\mathcal{Y} = D$. since $L_j(D) = b^j$ we have $\sum_j L_j(D) \omega_j = \omega$, hence $H(D) = e$.

The final witness $\mathbf{w}^*$ is obtained by the same linear combination as in PGL-Fold (Figure 2), with norm bounded by

$$\|\mathbf{w}^*\| \leq (b-1) + \sum_{j=1}^{k'-1} \|y_j - y_{j-1}\|_{\text{op}}(b-1) \leq (2T(k'-1) + 1)(b-1) \leq \gamma.$$

Soundness. Our extractor runs the extractor from the knowledge soundness of Theorem 2. We obtain relaxed witnesses $(\bar{\mathbf{w}}_j, s_j)_{j \in [0, k'-1]} \in (\Xi_{f, n, B_{\text{ext}}, \varrho}^{\text{acc}})^{k'}$. Instantiating the bounds of Theorem 2 with $k := k' - 1$ free challenges and output accumulator bounded by γ gives

$$B_{\text{ext}} = (2T)^{k'-1}((k' - 1)^2 + k')\gamma \quad \text{and} \quad \varrho = (2T)^{k'-1}.$$

Finally, under the hardness of $\text{MSIS}_{\kappa, m, 2\varrho B_{\text{ext}}}^{\infty, q}$, our extractor outputs *unique* $(\bar{\mathbf{w}} := \sum_{j=0}^{k'-1} b^j \bar{\mathbf{w}}_j, s_0)$.

Following the extraction of Theorem 2, for all $j \in [k' - 1]$ we have

$$\begin{aligned} \|s_j\|_{\text{op}} &\leq (2T)^{k'-j} \\ \|s_j \bar{\mathbf{w}}_j\| &\leq (2T)^{k'-j} 2\gamma' \\ \mathbf{A} \bar{\mathbf{w}}_j &= \mathbf{t}_j. \end{aligned}$$

For $j = 0$, we have

$$\begin{aligned} \|s_0\|_{\text{op}} &\leq (2T)^{k'-1} \\ \|s_0 \bar{\mathbf{w}}_j\| &\leq (2T)^{k'-1} 2\gamma' \\ \mathbf{A} \bar{\mathbf{w}}_0 &= \mathbf{t}_0. \end{aligned}$$

It follows that

$$\begin{aligned} \|s_0 \bar{\mathbf{w}}\| &= \|s_0 \sum_{j=0}^{k'-1} b^j \bar{\mathbf{w}}_j\| \\ &= \left\| \sum_{j=0}^{k'-1} b^j \frac{s_0}{s_j} s_j \bar{\mathbf{w}}_j \right\| \\ &\leq \left\| \sum_{j=1}^{k'-1} b^j (2T)^j (2T)^{k'-j} (2\gamma') \right\| + \|(2T)^{k'-1} (2\gamma')\| \\ &\leq k' \gamma (2T)^{k'-1} (2\gamma') \\ &\leq \gamma_{\text{ext}}. \end{aligned}$$

For the commitment of $\bar{\mathbf{w}}$, we have

$$\begin{aligned} \mathbf{A} \bar{\mathbf{w}} &= \mathbf{A} \left(\sum_{j=0}^{k'-1} b^j \bar{\mathbf{w}}_j \right) \\ &= \sum_{j=0}^{k'-1} b^j \mathbf{A} \bar{\mathbf{w}}_j = \sum_{j=0}^{k'-1} b^j \mathbf{t}_j = \mathbf{t}, \end{aligned}$$

where the last equality is the verifier's first check.

We now argue that $\Phi(X) := \sum_{i=1}^n \text{pow}_i(\beta) f_i(X)$ is such that $\Phi(\bar{\omega}_j) = e_j$. Similarly as in the proof of Theorem 2, combining the verification equation and Theorem 1, we create a polynomial

$$\Psi(\mathcal{Y}) := \sum_{r,s} (K_{rs}(\mathcal{Y}) - Q_{rs}(\mathcal{Y})) Z_{rs}(\mathcal{Y}) + \sum_{j=0}^{k'-1} L_j(\mathcal{Y}) (\Phi(\omega_j) - e_j).$$

Since the $\bar{\mathbf{w}}_j$'s are unique, the above identity holds for any accepting transcript at the evaluation point $\mathbf{y}$. We apply the Schwartz-Zippel over rings Lemma 1 and deduce that if Ψ were non-zero, then

$$\mathbb{P}_{\mathcal{Y} \leftarrow \mathcal{C}^{k'-1}}(\Psi(\mathcal{Y}) = 0) \leq \frac{d(k' - 1)}{|\mathcal{C}|},$$

the verifier sampling $y_1, \dots, y_{k'-1}$. Hence with probability at least $1 - \frac{d(k'-1)}{|\mathcal{C}|}$ we have $\Psi = 0$ as a polynomial, and by the uniqueness of the decomposition in a reduced Gröbner basis Proposition 5, we have that $(K_{rs})_{rs} = (Q_{rs})_{rs}$ up to syzygies and for all $j \in [0, k' - 1]$, $\Phi(\bar{\mathbf{w}}_j) = e_j$. In particular $P := \sum_{r,s} K_{rs} Z_{rs}$ is the quotient part of H , i.e.

$$H(\mathcal{Y}) = \sum_{j=0}^{k'-1} L_j(\mathcal{Y}) e_j + P(\mathcal{Y}),$$

which is the decomposition the prover computes in Figure 3. We deduce from the check $\sum_{j=0}^{k'-1} b^j e_j \stackrel{?}{=} e + \sum_{rs} Z_{rs}(D) K_{rs}(D)$ that

$$e = \sum_{j=0}^{k'-1} L_j(D) e_j + P(D).$$

It follows that $\Phi(\bar{\omega}) = e$. Via Theorem 1, $\|s_0\|_{\text{op}} \leq \varrho$ and $\|s_0 \bar{\mathbf{w}}\| \leq \gamma_{\text{ext}}$, which completes the proof.